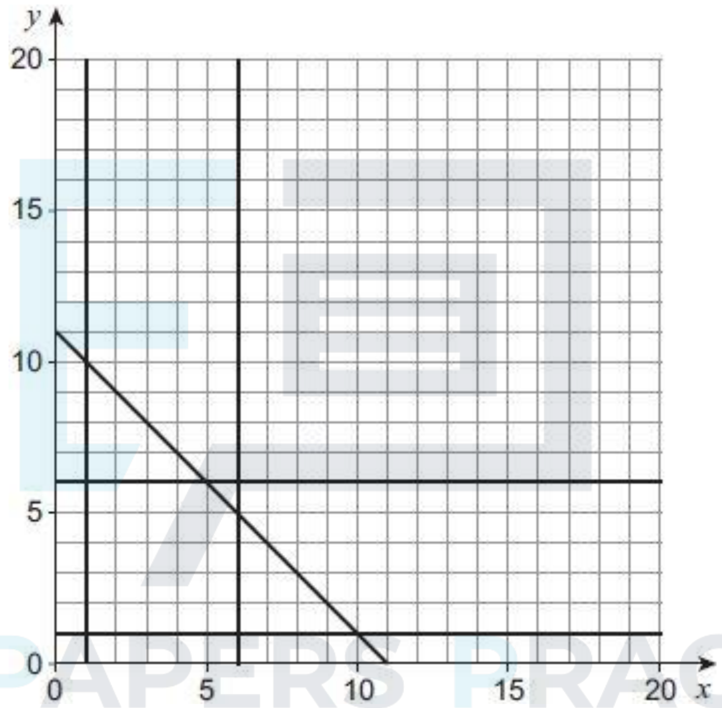
Q1.

A linear programming problem has the constraints

$$1 \le x \le 6$$
$$1 \le y \le 6$$
$$y \ge x$$
$$x + y \le 11$$

(a) (i) Complete **Figure 1** to identify the feasible region for the problem.

Figure 1



(2)

(ii) Determine the maximum value of $5x + 4y$ subject to the constraints.

(2)

(b) The simple-connected graph G has seven vertices.

The vertices of G have degree 1, 2, 3, v, w, x and y

(i) Explain why $x \ge 1$ and $y \ge 1$

(1)

(ii) Explain why $x \le 6$ and $y \le 6$

(1)

(iii) Explain why $x + y \le 11$.

(2)

(iv) State an additional constraint that applies to the values of x and y in this context.

(1)

(c) The graph G also has eight edges. The inequalities used in part (a)(i) apply to the graph G.

(i) Given that $v + w = 4$, find all the feasible values of x and y.

(3)

(ii) It is also given that the graph G is semi-Eulerian.

On **Figure 2**, draw G.

Figure 2

(2)
(Total 14 marks)

Q2.

A family business makes and sells two kinds of kitchen table.

Each pine table takes 6 hours to make and the cost of materials is £30.
Each oak table takes 10 hours to make and the cost of materials is £70.
Each month, the business has 360 hours available for making the tables and £2100 available for the materials.
Each month, the business sells all of its tables to a wholesaler.

The wholesaler specifies that it requires at least 10 oak tables per month and at least as many pine tables as oak tables.

Each pine table sold gives the business a profit of £40 and each oak table sold gives the business a profit of £75.

Use a graphical method to find the number of each type of table the business should make each month, in order to maximise its total profit.

Show clearly how you obtain your answer.

(Total 8 marks)

Q3.

The feasible region of a linear programming problem is defined by

$$x + y \leq 60$$

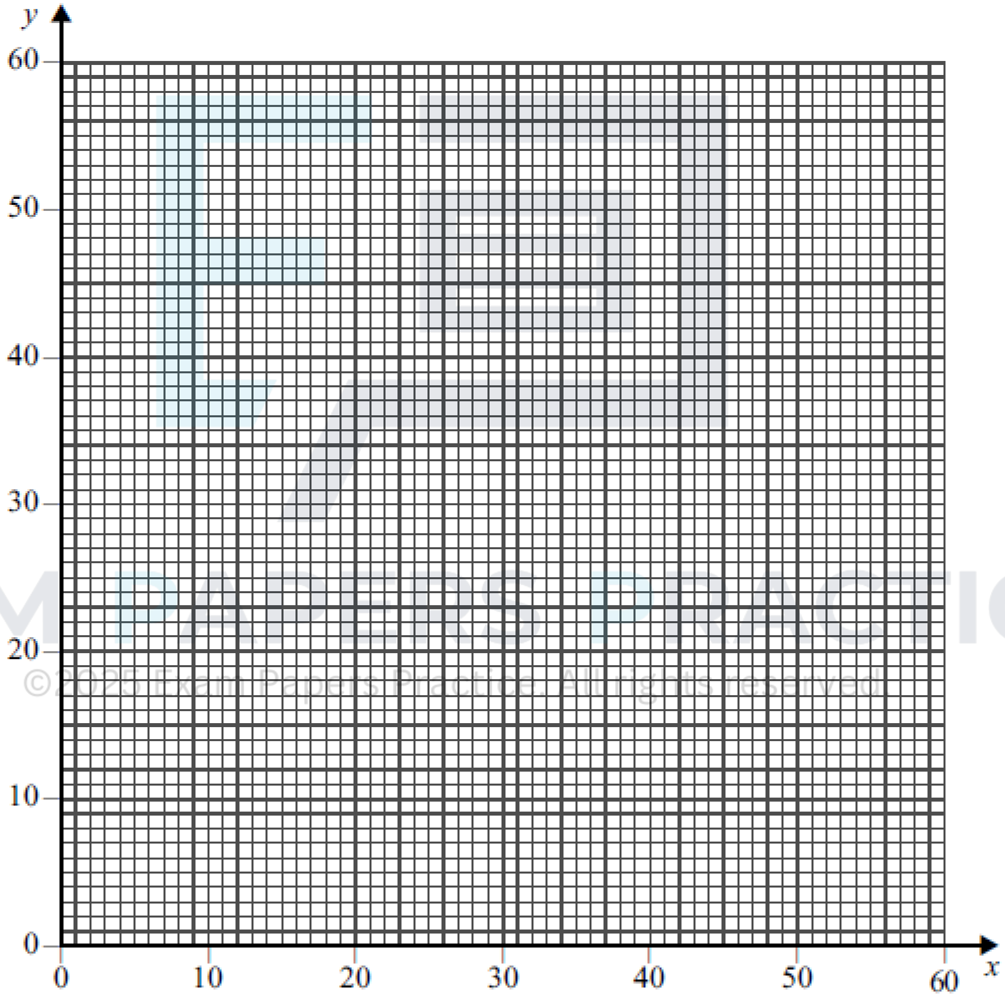
$$2x + y \leq 80$$

$$y \geq 20$$

$$x \geq 15$$

$$y \geq x$$

(a) On the grid below, draw a suitable diagram to represent these inequalities and indicate the feasible region.



(5)

(b) In each of the following cases, use your diagram to find the maximum value of P on the feasible region. In each case, state the corresponding values of x and y.

(i) $P = x + 4y$

(2)

(ii) $P = 4x + y$

(3)

(Total 10 marks)

Q4.

Paul is a florist. Every day, he makes three types of floral bouquet: gold, silver and bronze.

Each gold bouquet has 6 roses, 6 carnations and 6 dahlias.

Each silver bouquet has 4 roses, 6 carnations and 4 dahlias.

Each bronze bouquet has 3 roses, 4 carnations and 4 dahlias.

Each day, Paul must use at least 420 roses and at least 480 carnations, but he can use at most 720 dahlias.

Each day, Paul makes x gold bouquets, y silver bouquets and z bronze bouquets.

(a) In addition to $x \geq 0$, $y \geq 0$ and $z \geq 0$, find three inequalities in x, y and z that model the above constraints.

(3)

(b) On a particular day, Paul makes the same number of silver bouquets as bronze bouquets.

(i) Show that x and y must satisfy the following inequalities.

$$6x + 7y \geq 420$$

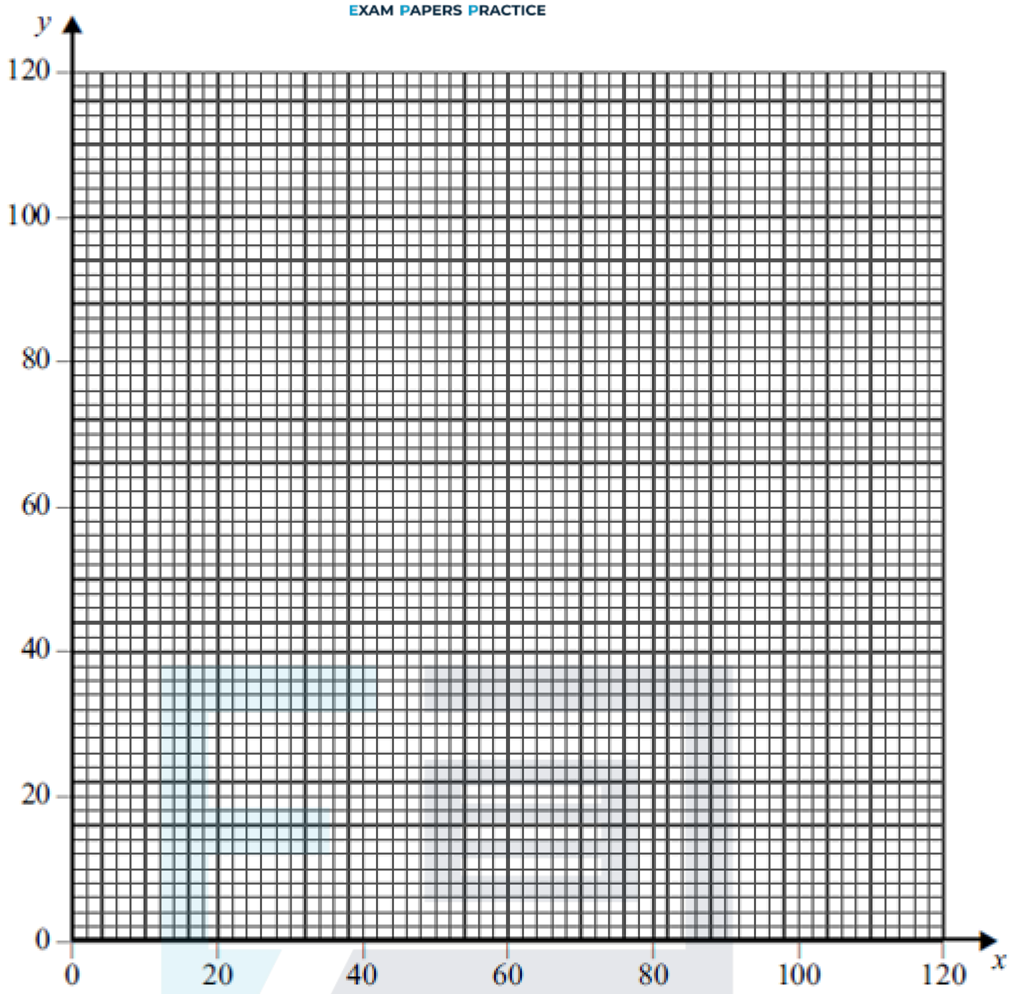
$$3x + 5y \geq 240$$

$$3x + 4y \leq 360$$

(2)

(ii) Paul makes a profit of £4 on each gold bouquet sold, a profit of £2.50 on each silver bouquet sold and a profit of £2.50 on each bronze bouquet sold. Each day, Paul sells all the bouquets he makes. Paul wishes to maximise his daily profit, £P.

Draw a suitable diagram, on the grid below, to enable this problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(6)

(iii) Use your diagram to find Paul's maximum daily profit and the number of each type of bouquet he must make to achieve this maximum.

(2)

(c) On another day, Paul again makes the same number of silver bouquets as bronze bouquets, but he makes a profit of £2 on each gold bouquet sold, a profit of £6 on each silver bouquet sold and a profit of £6 on each bronze bouquet sold.

Find Paul's maximum daily profit, and the number of each type of bouquet he must make to achieve this maximum.

(3)

(Total 16 marks)

Q5.

The feasible region of a linear programming problem is determined by the following:

$$y \geq 20$$

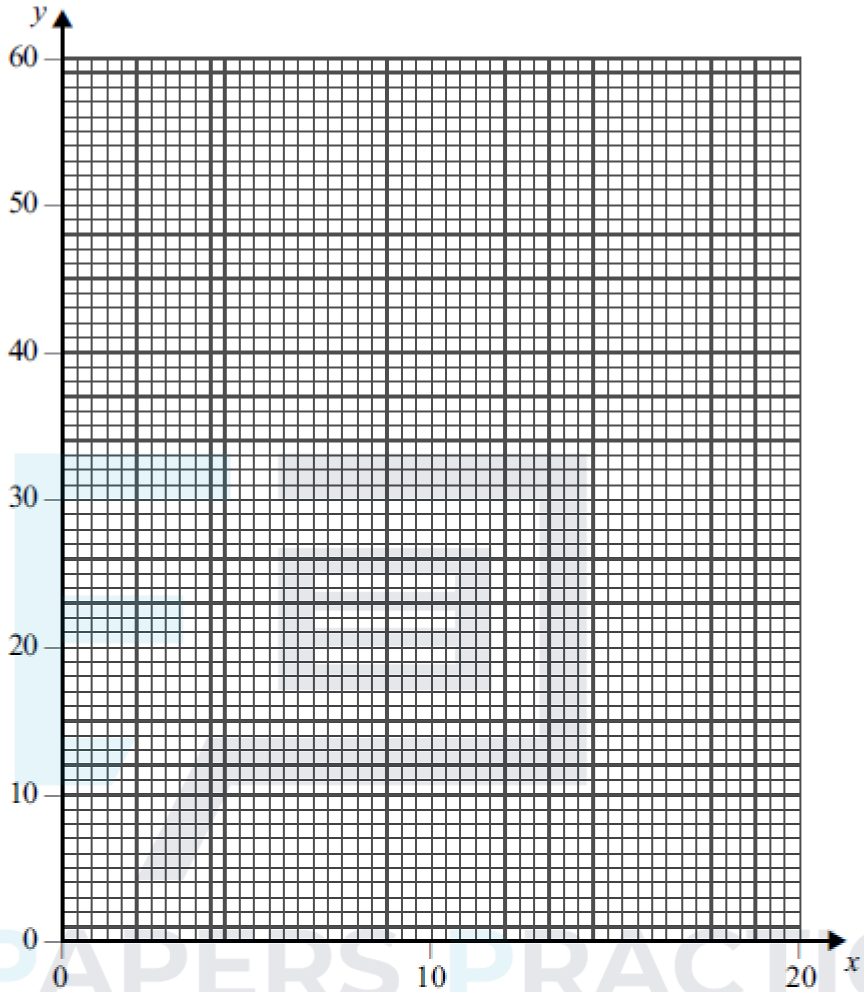
$$x + y \geq 25$$

$$5x + 2y \leq 100$$

$$y \leq 4x$$

$$y \geq 2x$$

(a) On the **figure** below, draw a suitable diagram to represent the inequalities and indicate the feasible region.



(6)

(b) Use your diagram to find the minimum value of P, on the feasible region, in the case where:

(i) $P = x + 2y$;

(ii) $P = -x + y$.

In each case, state the corresponding values of x and y.

(4)
(Total 10 marks)

Q6.

Ollyin is buying new pillows for his hotel. He buys three types of pillow: soft, medium and firm.

He must buy at least 100 soft pillows and at least 200 medium pillows.

He must buy at least 400 pillows in total.

Soft pillows cost £4 each. Medium pillows cost £3 each. Firm pillows cost £4 each.

He wishes to spend no more than £1800 on new pillows.

At least 40% of the new pillows must be medium pillows.

Ollyin buys x soft pillows, y medium pillows and z firm pillows.

(a) In addition to $x \geq 0$, $y \geq 0$ and $z \geq 0$, find five inequalities in x, y and z that model the above constraints.

(3)

(b) Ollyin decides to buy twice as many soft pillows as firm pillows.

(i) Show that three of your answers in part (a) become

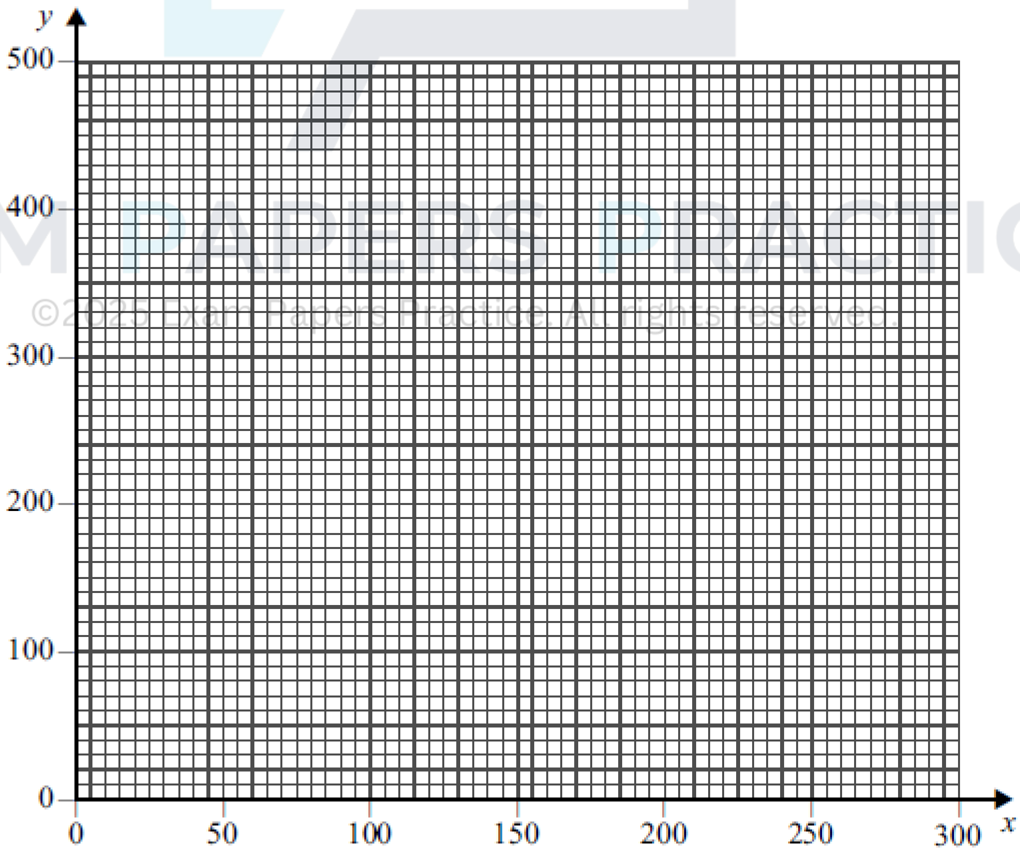
$$3x + 2y \geq 800$$

$$2x + y \leq 600$$

$$y \geq x$$

(3)

(ii) On the grid below, draw a suitable diagram to represent Ollyin's situation, indicating the feasible region.



(5)

(iii) Use your diagram to find the maximum total number of pillows that Ollyin can buy.

(2)

(iv) Find the number of each type of pillow that Ollyin can buy that corresponds to your answer to part (b)(iii).

(1)

(Total 14 marks)

Q7.

Herman is packing some hampers. Each day, he packs three types of hamper: basic, standard and luxury.

Each basic hamper has 6 tins, 9 packets and 6 bottles.

Each standard hamper has 9 tins, 6 packets and 12 bottles.

Each luxury hamper has 9 tins, 9 packets and 18 bottles.

Each day, Herman has 600 tins and 600 packets available, and he must use at least 480 bottles.

Each day, Herman packs x basic hampers, y standard hampers and z luxury hampers.

(a) In addition to $x \geq 0$, $y \geq 0$ and $z \geq 0$, find three inequalities in x, y and z that model the above constraints, simplifying each inequality.

(4)

(b) On a particular day, Herman packs the same number of standard hampers as luxury hampers.

(i) Show that your answers in part (a) become

$$x + 3y \leq 100$$

$$3x + 5y \leq 200$$

$$x + 5y \geq 80$$

(2)

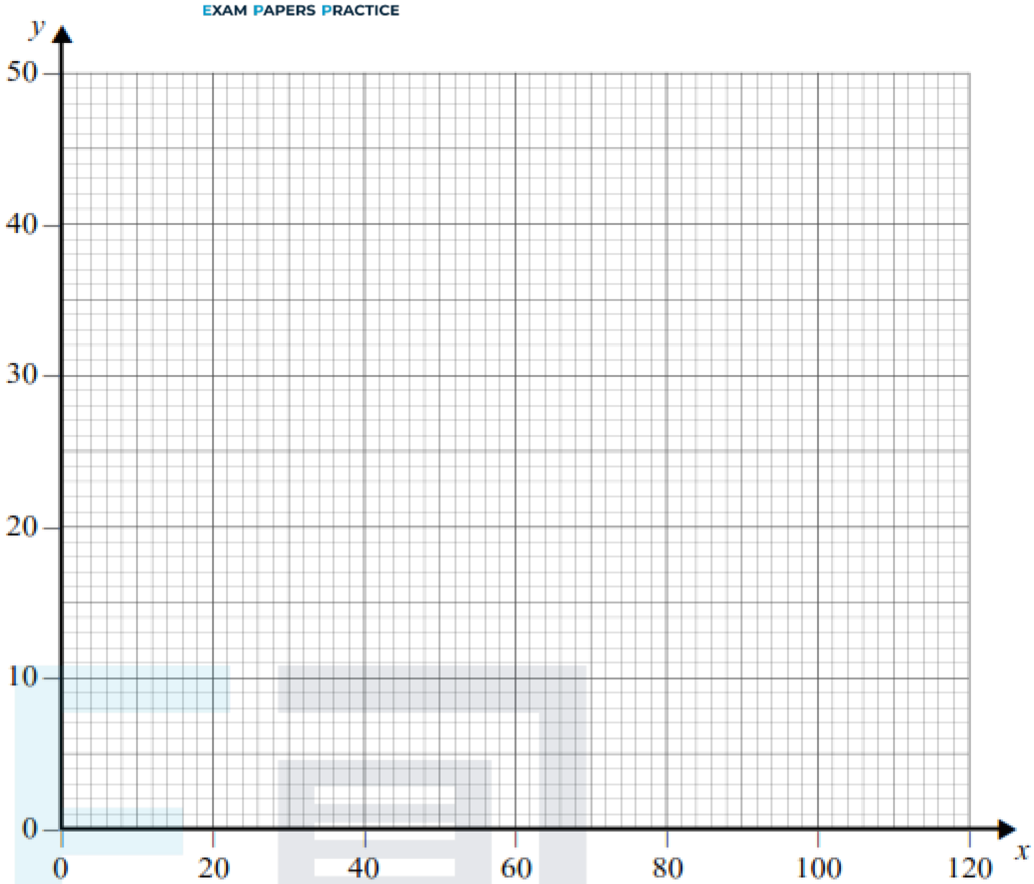
(ii) On the grid below, draw a suitable diagram to represent Herman's situation, indicating the feasible region.

(4)

(iii) Use your diagram to find the maximum total number of hampers that Herman can pack on that day.

(2)

(iv) Find the number of each type of hamper that Herman packs that corresponds to your answer to part (b)(iii).



(1)

(Total 13 marks)

Q8.

A builder needs some screws, nails and plugs. At the local store, there are packs containing a mixture of the three items.

A DIY pack contains 200 nails, 200 screws and 100 plugs.

A trade pack contains 1000 nails, 1500 screws and 2500 plugs.

A DIY pack costs £2.50 and a trade pack costs £15.

The builder needs at least 5000 nails, 6000 screws and 4000 plugs.

The builder buys x DIY packs and y trade packs and wishes to keep his total cost to a minimum.

(a) Formulate the builder's situation as a linear programming problem.

(4)

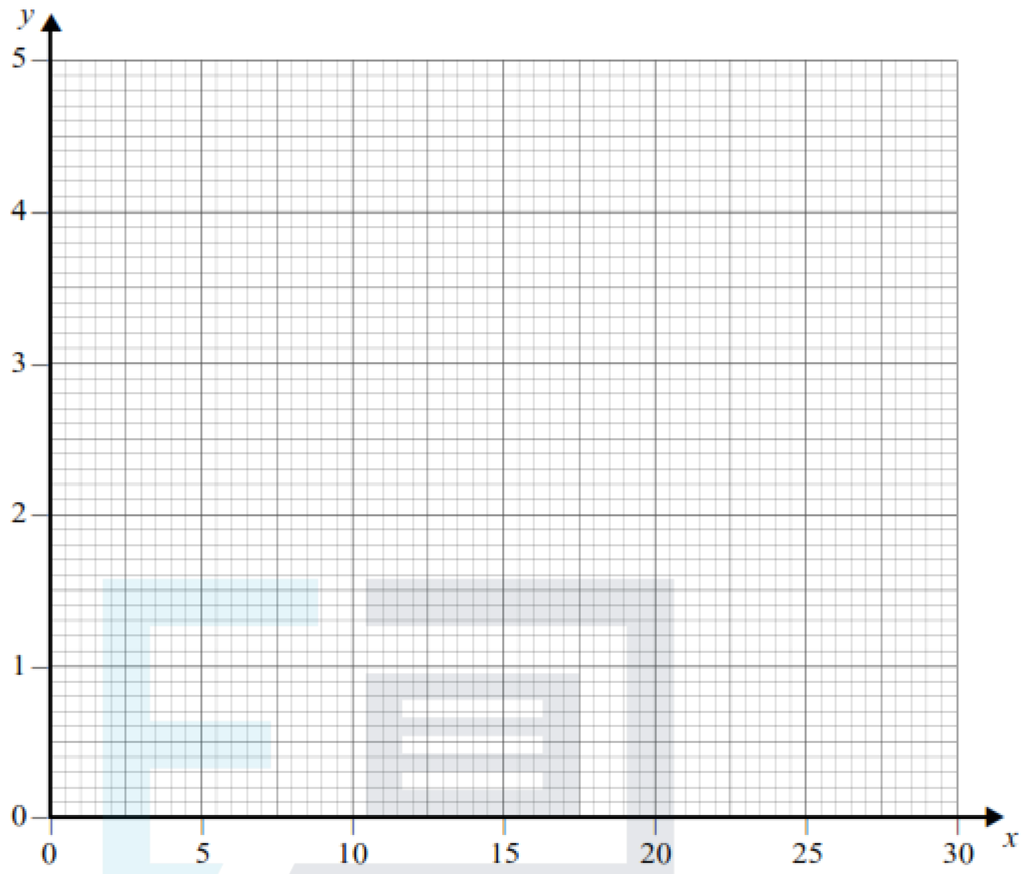
(b) (i) On the grid below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of an objective line.

(6)

(ii) Use your diagram to find the number of each type of pack that the builder should buy in order to minimise his cost.

(1)

(iii) Find the builder's minimum cost.



(1)

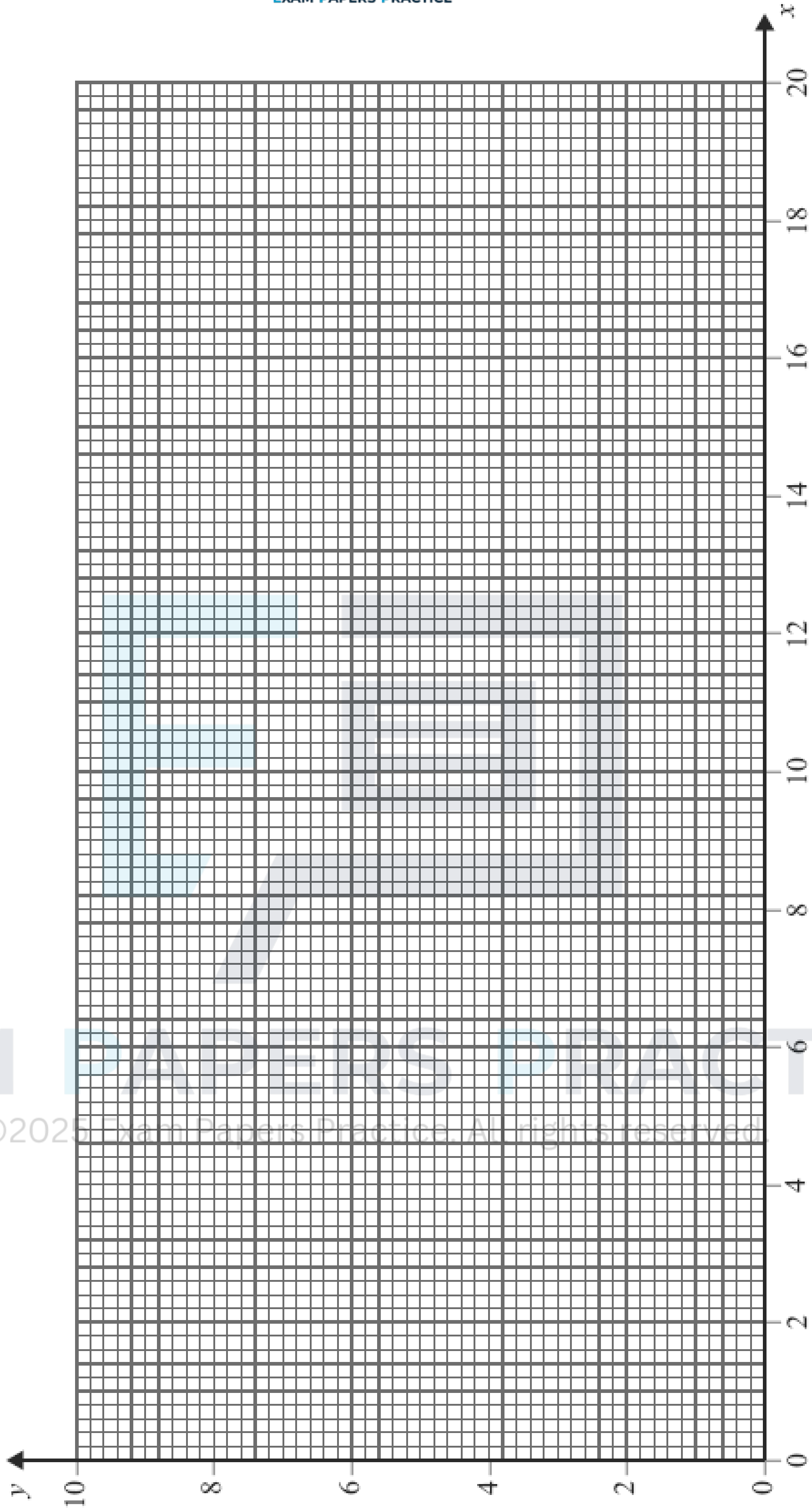
(Total 12 marks)

Q9.

The feasible region of a linear programming problem is represented by the following:

$$x \geq 0,\ y \geq 0$$
$$x + 4y \leq 36$$
$$4x + y \leq 68$$
$$y \leq 2x$$
$$y \geq \frac{1}{4}x$$

(a) On the figure below, draw a suitable diagram to represent the inequalities and indicate the feasible region.



(6)

(b) Use your diagram to find the maximum value of P, stating the corresponding coordinates, on the feasible region, in the case where:

(i) $P = x + 5y$;

(2)

(ii) $P = 5x + y$.

(2)
(Total 10 marks)

Q10.

Phil is to buy some squash balls for his club. There are three different types of ball that he can buy: slow, medium and fast.

He must buy at least 190 slow balls, at least 50 medium balls and at least 50 fast balls.

He must buy at least 300 balls in total.

Each slow ball costs £2.50, each medium ball costs £2.00 and each fast ball costs £2.00.

He must spend no more than £1000 in total.

At least 60% of the balls that he buys must be slow balls.

Phil buys x slow balls, y medium balls and z fast balls.

(a) Find six inequalities that model Phil's situation.

(4)

(b) Phil decides to buy the same number of medium balls as fast balls.

(i) Show that the inequalities found in part (a) simplify to give

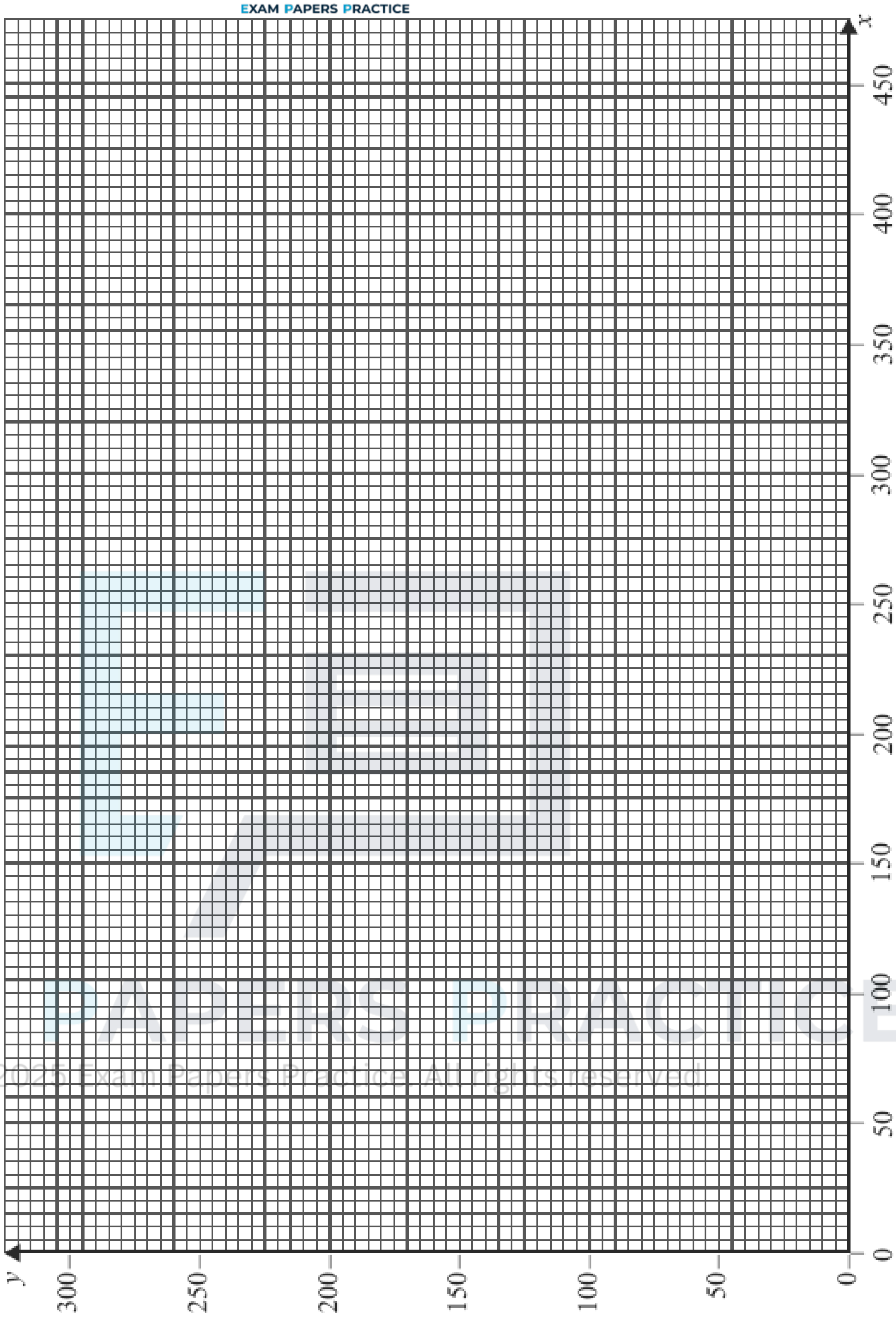
$$x \geq 190,\ y \geq 50,\ x + 2y \geq 300,\ 5x + 8y \leq 2000,\ y \leq \frac{1}{3}x$$

(2)

(ii) Phil sells all the balls that he buys to members of the club. He sells each slow ball for £3.00, each medium ball for £2.25 and each fast ball for £2.25. He wishes to maximise his profit.

On the figure below, draw a diagram to enable this problem to be solved graphically, indicating the feasible region and the direction of an objective line.

$$x \geq 190,\ y \geq 50,\ x + 2y \geq 300,\ 5x + 8y \leq 2000,\ y \leq \frac{1}{3}x$$



(7)

(iii) Find Phil's maximum possible profit and state the number of each type of ball that he must buy to obtain this maximum profit.

(4)

(Total 17 marks)

Q11.

Each year, farmer Giles buys some goats, pigs and sheep.

He must buy at least 110 animals.
He must buy at least as many pigs as goats.
The total of the number of pigs and the number of sheep that he buys must not be greater than 150.
Each goat costs £16, each pig costs £8 and each sheep costs £24.
He has £3120 to spend on the animals.

At the end of the year, Giles sells all of the animals. He makes a profit of £70 on each goat, £30 on each pig and £50 on each sheep. Giles wishes to maximize his total profit, £P.

Each year, Giles buys x goats, y pigs and z sheep.

(a) Formulate Giles's situation as a linear programming problem.

(5)

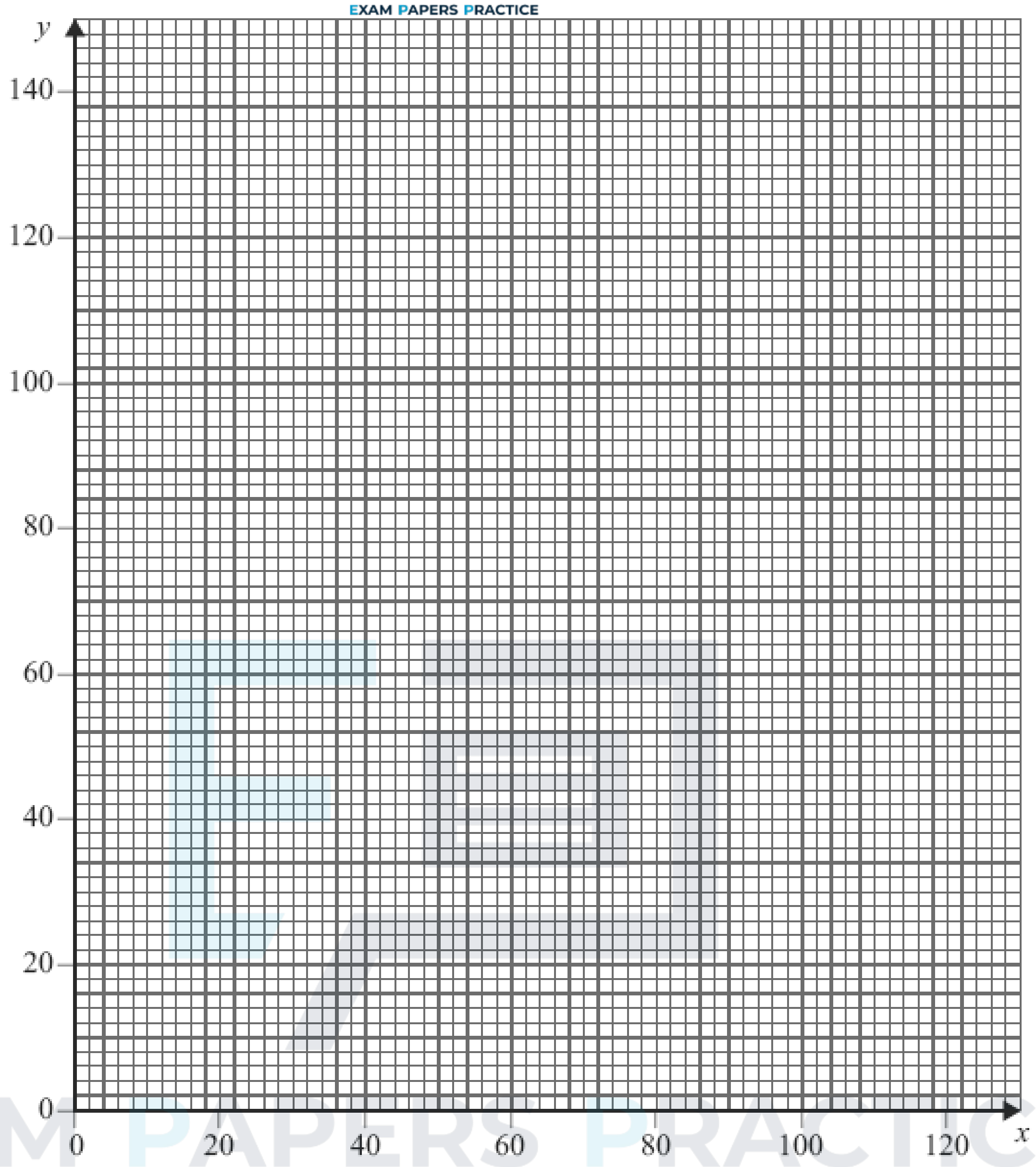
(b) One year, Giles buys 30 sheep.

(i) Show that the constraints for Giles's situation for this year can be modelled by

$$y \geq x, \quad 2x + y \leq 300, \quad x + y \geq 80, \quad y \leq 120$$

(2)

(ii) On the diagram below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(8)

(iii) Find Giles's maximum profit for this year and the number of each animal that he must buy to obtain this maximum profit.

(3)
(Total 18 marks)

Q12.

Each day, a factory makes three types of widget: basic, standard and luxury. The widgets produced need three different components: type A, type B and type C.

Basic widgets need 6 components of type A, 6 components of type B and 12 components of type C.
Standard widgets need 4 components of type A, 3 components of type B and 18 components of type C.
Luxury widgets need 2 components of type A, 9 components of type B and 6 components of type C.
Each day, there are 240 components of type A available, 300 of type B and

900 of type C.

Each day, the factory must use at least twice as many components of type C as type B.

Each day, the factory makes x basic widgets, y standard widgets and z luxury widgets.

(a) In addition to $x \geq 0$, $y \geq 0$ and $z \geq 0$, find four inequalities in x, y and z that model the above constraints, simplifying each inequality.

(8)

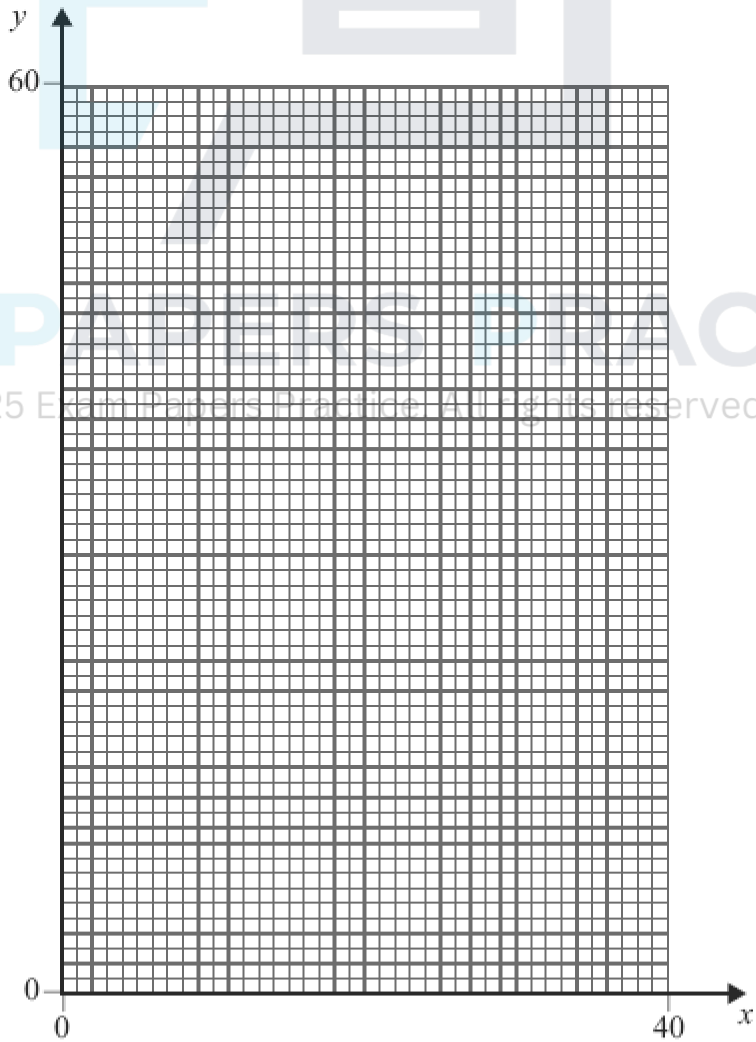
(b) Each day, the factory makes the maximum possible number of widgets. On a particular day, the factory must make the same number of luxury widgets as basic widgets.

(i) Show that your answers in part (a) become

$$2x + y \leq 60, \quad 5x + y \leq 100, \quad x + y \leq 50, \quad y \geq x$$

(3)

(ii) Using the axes below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region.



(5)

(iii) Find the total number of widgets made on that day.

(2)

(iv) Find all possible combinations of the number of each type of widget made that correspond to this maximum number.

(3)

(Total 21 marks)

Q13.

The feasible region of a linear programming problem is represented by

$$x + y \leq 30$$

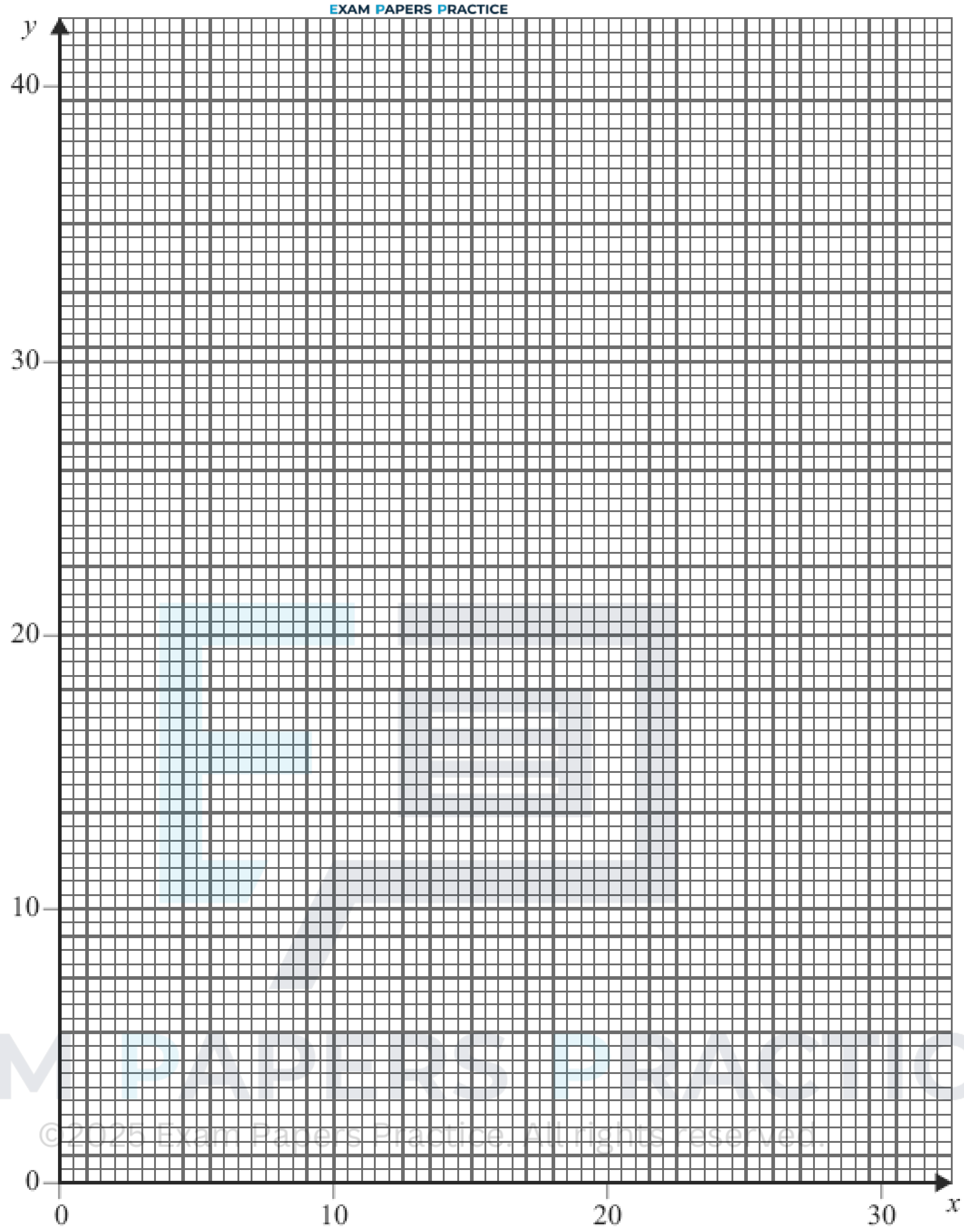
$$2x + y \leq 40$$

$$y \geq 5$$

$$x \geq 4$$

$$y \geq \frac{1}{2}x$$

(a) On the figure below, draw a suitable diagram to represent these inequalities and indicate the feasible region.



(5)

(b) Use your diagram to find the maximum value of F, on the feasible region, in the case where:

(i) $F = 3x + y$;

(2)

(ii) $F = x + 3y$.

(2)

(Total 9 marks)

Q14.

Each day, a factory makes three types of hinge: basic, standard and luxury. The hinges produced need three different components: type A, type B and type C.

Basic hinges need 2 components of type A, 3 components of type B and 1 component of type C.

Standard hinges need 4 components of type A, 2 components of type B and 3 components of type C.

Luxury hinges need 3 components of type A, 4 components of type B and 5 components of type C.

Each day, there are 360 components of type A available, 270 of type B and 450 of type C.

Each day, the factory must use at least 720 components in total.

Each day, the factory must use at least 40% of the total components as type A.

Each day, the factory makes x basic hinges, y standard hinges and z luxury hinges.

In addition to $x \geq 0$, $y \geq 0$, $z \geq 0$, find five inequalities, each involving x, y and z, which must be satisfied. Simplify each inequality where possible.

(Total 8 marks)

Q15.

A factory makes two types of lock, standard and large, on a particular day.

On that day:

the maximum number of standard locks that the factory can make is 100;
the maximum number of large locks that the factory can make is 80;
the factory must make at least 60 locks in total;
the factory must make more large locks than standard locks.

Each standard lock requires 2 screws and each large lock requires 8 screws, and on that day the factory must use at least 320 screws.

On that day, the factory makes x standard locks and y large locks.

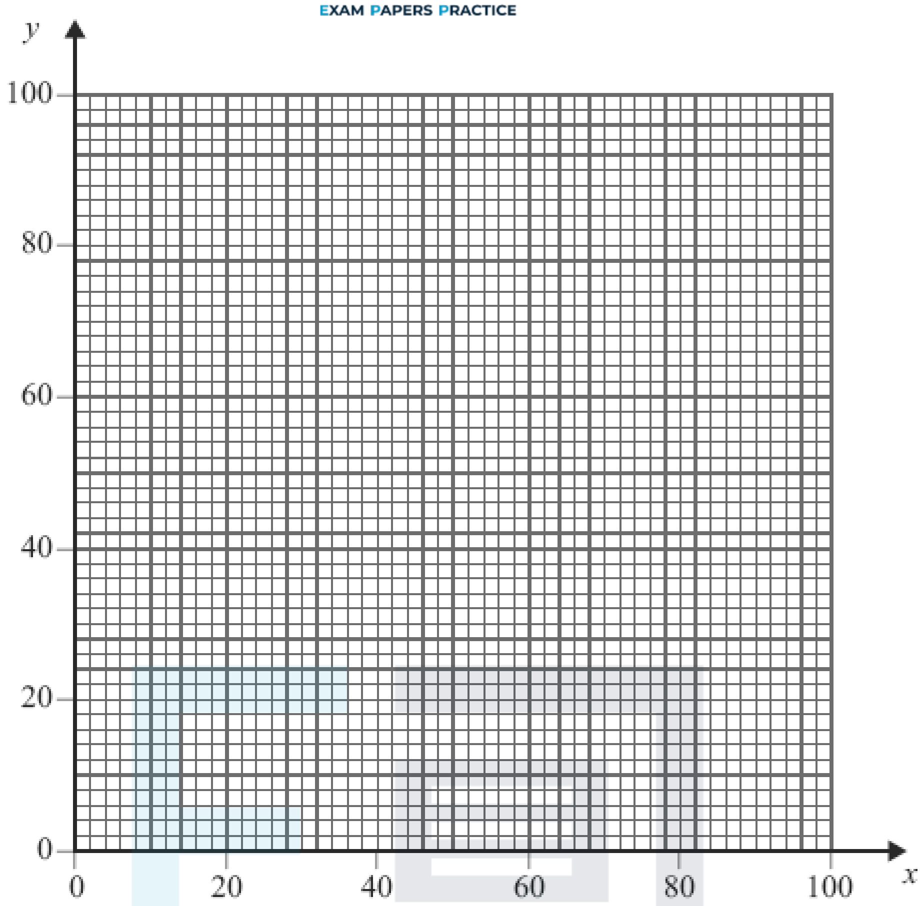
Each standard lock costs £1.50 to make and each large lock costs £3 to make.

The manager of the factory wishes to minimise the cost of making the locks.

(a) Formulate the manager's situation as a linear programming problem.

(5)

(b) On the figure below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(6)

(c) Find the values of x and y that correspond to the minimum cost. Hence find this minimum cost.

(4)

(Total 15 marks)

Q16.

Dino is to have a rectangular swimming pool at his villa.

He wants its width to be at least 2 metres and its length to be at least 5 metres.

He wants its length to be at least twice its width.

He wants its length to be no more than three times its width.

Each metre of the width of the pool costs £1000 and each metre of the length of the pool costs £500.

He has £9000 available.

Let the width of the pool be x metres and the length of the pool be y metres.

(a) Show that one of the constraints leads to the inequality

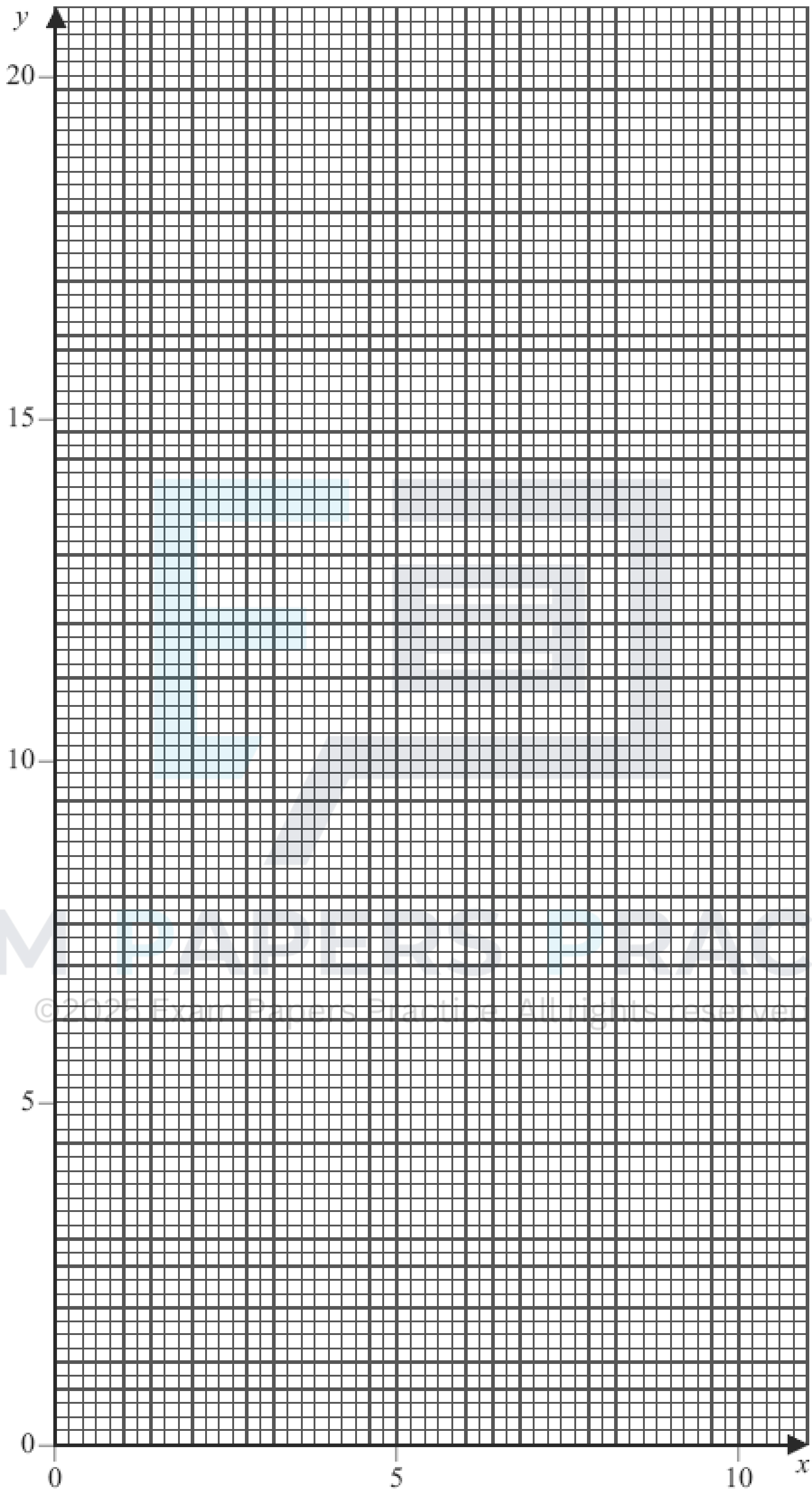
$$2x + y \leq 18$$

(1)

(b) Find four further inequalities.

(3)

(c) On the figure below, draw a suitable diagram to show the feasible region.



(6)

(d) Use your diagram to find the maximum width of the pool. State the corresponding length of the pool.

(3)
(Total 13 marks)

Q17.

The Jolly Company sells two types of party pack: excellent and luxury.

Each excellent pack has five balloons and each luxury pack has ten balloons.

Each excellent pack has 32 sweets and each luxury pack has 8 sweets.

The company has 1500 balloons and 4000 sweets available.

The company sells at least 50 of each type of pack and at least 140 packs in total.

The company sells x excellent packs and y luxury packs.

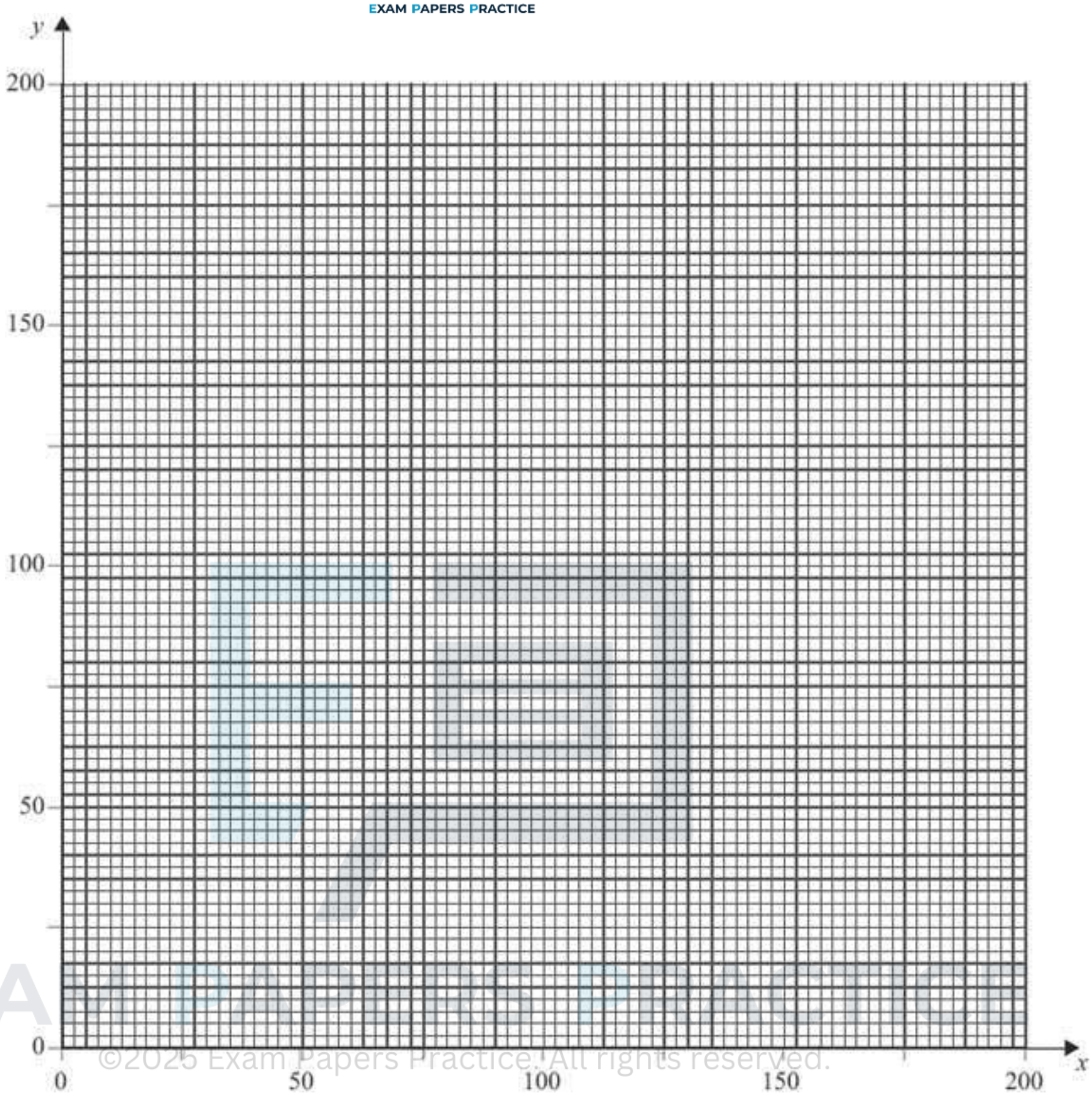
(a) Show that the above information can be modelled by the following inequalities.

$$x + 2y \leq 300, \quad 4x + y \leq 500, \quad x \geq 50, \quad y \geq 50, \quad x + y \geq 140$$

(4)

(b) The company sells each excellent pack for 80p and each luxury pack for £1.20. The company needs to find its minimum and maximum total income.

(i) On the figure below, draw a suitable diagram to enable this linear programming problem to be solved graphically, indicating the feasible region and an objective line.



(8)

(ii) Find the company's maximum total income and state the corresponding number of each type of pack that needs to be sold.

(2)

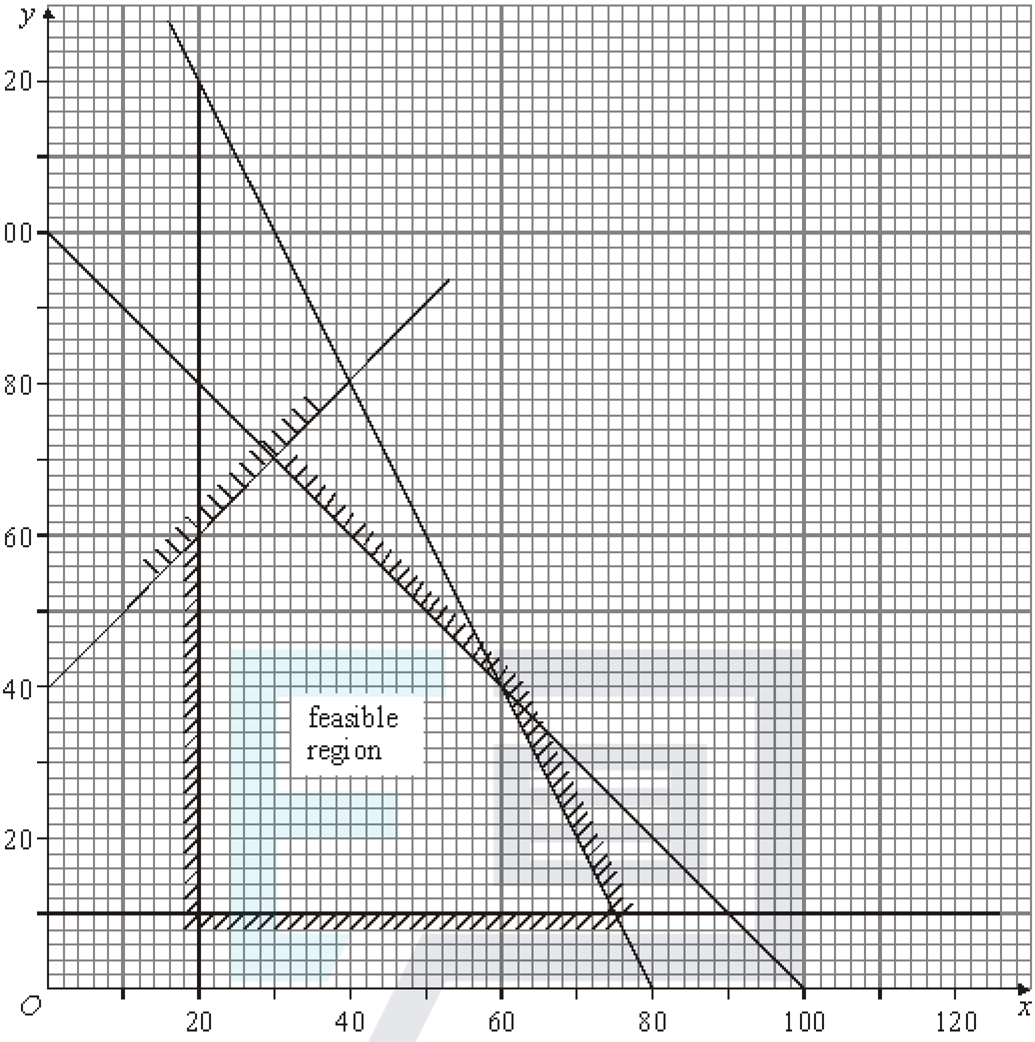
(iii) Find the company's minimum total income and state the corresponding number of each type of pack that needs to be sold.

(2)

(Total 16 marks)

Q18.

The diagram shows the feasible region of a linear programming problem.



(a) On the feasible region, find:

(i) the maximum value of $2x + 3y$;

(2)

(ii) the maximum value of $3x + 2y$;

(2)

(iii) the minimum value of $–2x + y$.

(2)

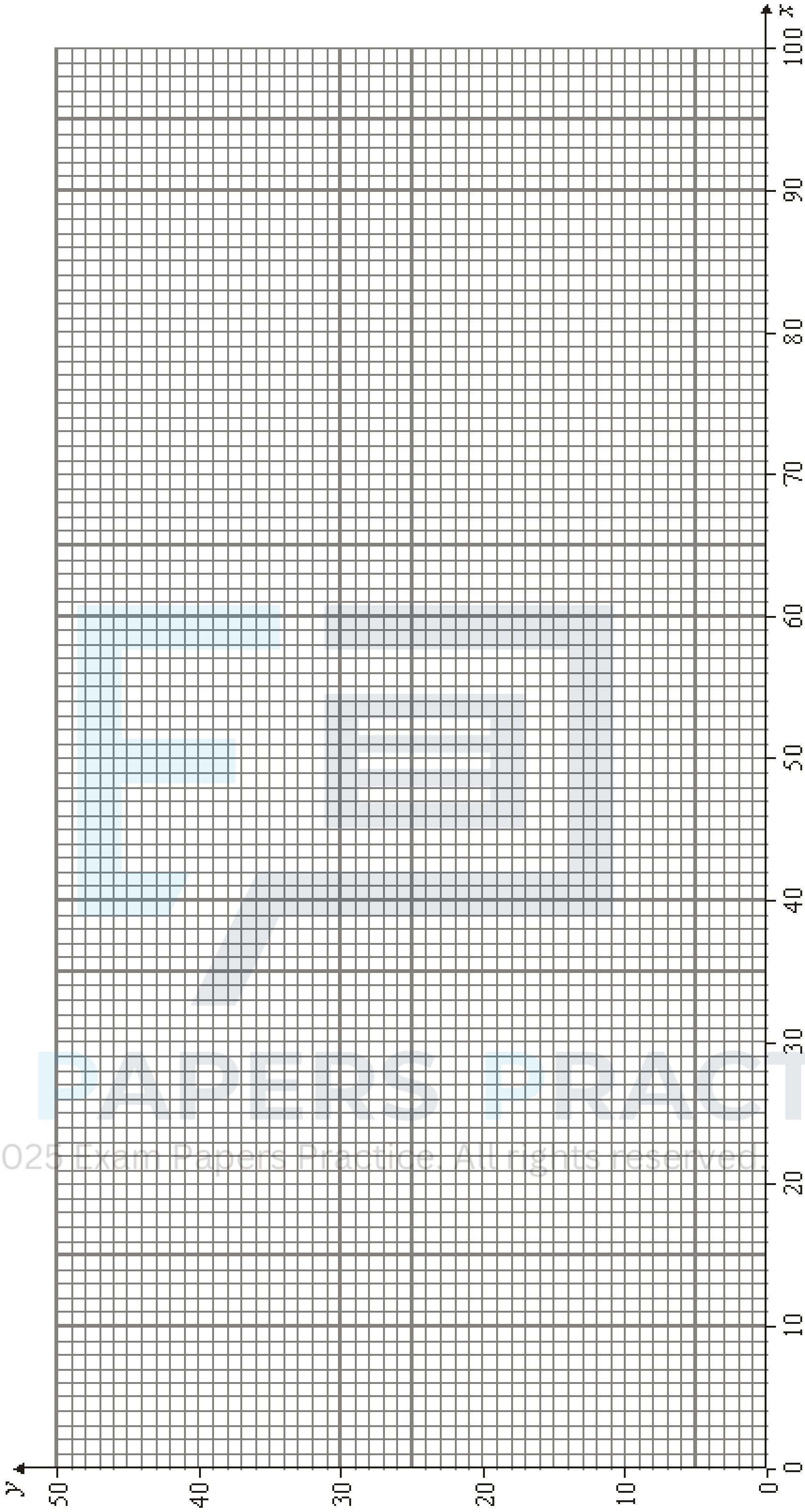
(b) Find the 5 inequalities that define the feasible region.

(6)

(Total 12 marks)

Q19.

The figure below, is provided for use in this question.
Ernesto is to plant a garden with two types of tree: palms and conifers.



He is to plant at least 10, but not more than 80 palms.

He is to plant at least 5, but not more than 40 conifers.

He cannot plant more than 100 trees in total.

Each palm needs 20 litres of water each day and each conifer needs 60 litres of water each day. There are 3000 litres of water available each day.

Ernesto makes a profit of £2 on each palm and £1 on each conifer that he plants and he wishes to maximise his profit.

Ernesto plants x palms and y conifers.

(a) Formulate Ernesto's situation as a linear programming problem.

(5)

(b) On the figure below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.

(7)

(c) Find the maximum profit for Ernesto.

(2)

(d) Ernesto introduces a new pricing structure in which he makes a profit of £1 on each palm and £4 on each conifer.

Find Ernesto's new maximum profit and the number of each type of tree that he should plant to obtain this maximum profit.

(2)
(Total 16 marks)

Q20.

A bakery makes two types of pizza, large and medium.

Every day the bakery must make at least 40 of each type.

Every day the bakery must make at least 120 in total but not more than 400 pizzas in total.

Each large pizza takes 4 minutes to make, and each medium pizza takes 2 minutes to make. There are four workers available, each for five hours a day, to make the pizzas.

The bakery makes a profit of £3 on each large pizza sold and £1 on each medium pizza sold.

Each day, the bakery makes and sells x large pizzas and y medium pizzas.

The bakery wishes to maximise its profit, £P.

(a) Show that one of the constraints leads to the inequality

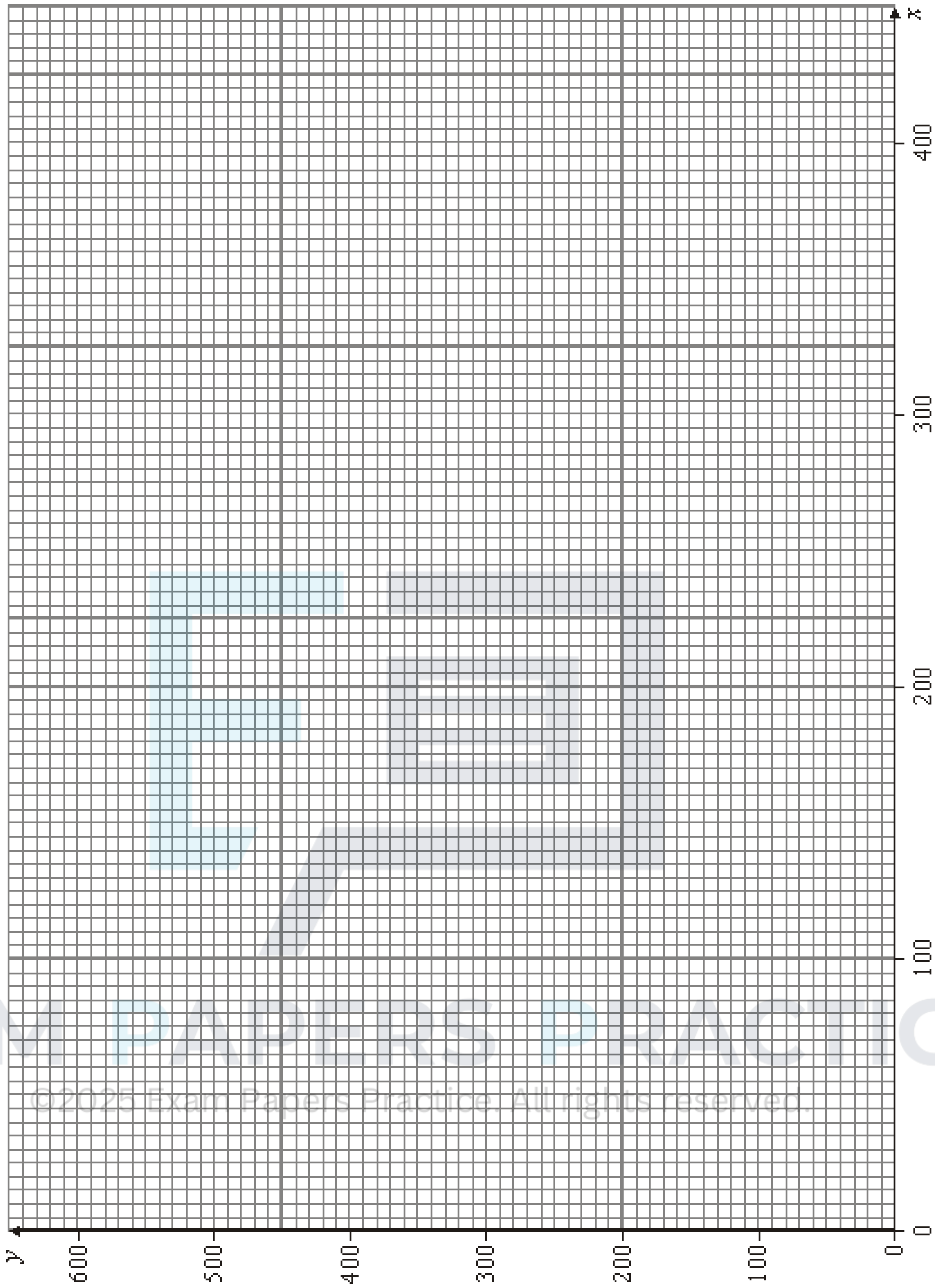
$$2x + y \leq 600$$

(1)

(b) Formulate this situation as a linear programming problem.

(3)

(c) On the figure below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and an objective line.



(5)

(d) Use your diagram to find the maximum daily profit.

(2)

(e) The bakery introduces a new pricing structure in which the profit is £2 on each large pizza sold and £2 on each medium pizza sold.

(i) Find the new maximum daily profit for the bakery.

(3)

(ii) Write down the number of different combinations that would give the new maximum daily profit.

(2)
(Total 16 marks)

Q21.

A company makes two types of boxes of chocolates, executive and luxury.

Every hour the company must make at least 15 of each type and at least 35 in total.

Each executive box contains 20 dark chocolates and 12 milky chocolates.

Each luxury box contains 10 dark chocolates and 18 milky chocolates.

Every hour the company has 600 dark chocolates and 600 milky chocolates available.

The company makes a profit of £1.50 on each executive box and £1 on each luxury box.

The company makes and sells x executive boxes and y luxury boxes every hour.

The company wishes to maximise its hourly profit, £P.

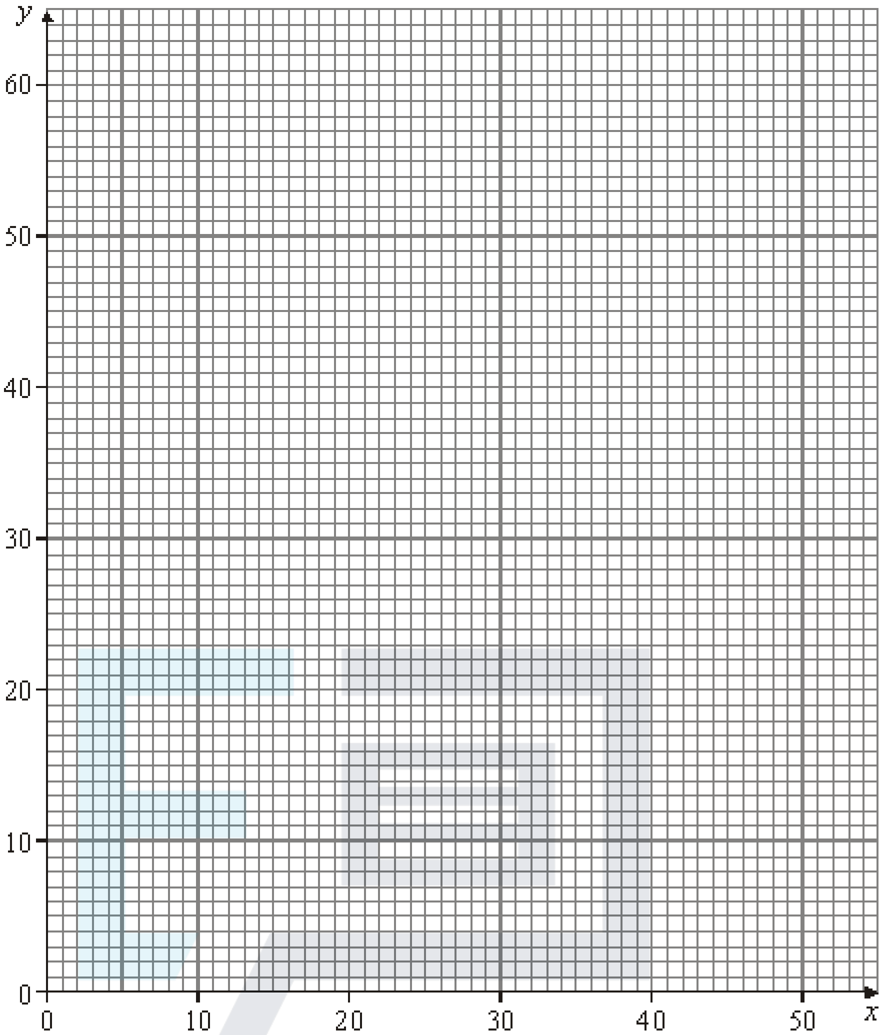
(a) Show that one of the constraints leads to the inequality $2x + 3y \leq 100$.

(1)

(b) Formulate the company's situation as a linear programming problem.

(4)

(c) On the figure below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and an objective line.



(6)

(d) Use your diagram to find the maximum hourly profit.

(2)
(Total 13 marks)

Q22.

A hotel manageress is to order some sunbeds. There are two types available: wooden and plastic.

She must order at least 30 wooden sunbeds and at least 60 plastic sunbeds.

She can order at most 200 sunbeds.

Each wooden sunbed costs £40 and occupies an area of 2 square metres.

Each plastic sunbed costs £12 and occupies an area of 3 square metres.

She has £3600 to spend and the sunbeds must occupy an area of at least 300 square metres.

She must order at least 50% more plastic sunbeds than wooden sunbeds.

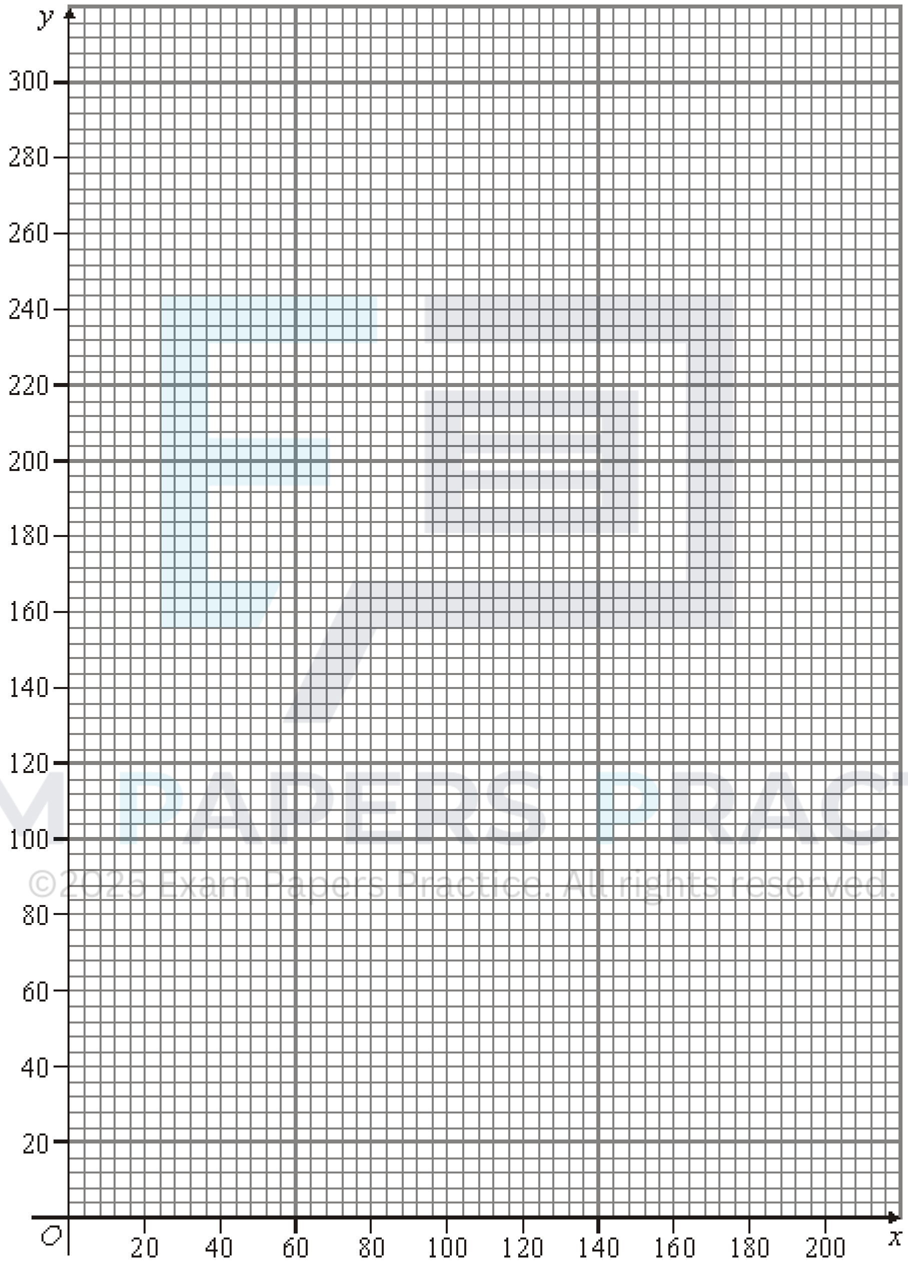
The manageress orders x wooden sunbeds and y plastic sunbeds.

(a) State, giving a reason in each case, why the manageress's situation can be modelled by the following inequalities:

$x \geq 30$, $y \geq 60$, $x + y \leq 200$, $2x + 3y \geq 300$, $10x + 3y \leq 900$ and $2y \geq 3x$.

(4)

(b) On the figure below, draw a suitable diagram to illustrate the situation graphically, indicating the feasible region



(6)

(c) State which inequality does **not** affect the feasible region.

(1)

(d) The hotel is awarded points on the number of sunbeds available for guests. Each wooden sunbed gains four points and each plastic sunbed gains five points.

By using an objective line, or otherwise, find the maximum number of points the hotel can be awarded and the number of each type of sunbed that corresponds to this maximum

(3)

(Total 14 marks)

Q23.

A small company makes two different types of boxes of Christmas crackers: luxury and standard.

Each hour, the company must make at least five boxes of each type, but not more than 25 standard boxes.

Each luxury box requires 20 toys, and each standard box 10 toys.

There are 600 toys available each hour.

The company must make at least half as many standard boxes as luxury boxes.

Each luxury box is sold at a profit of £3 and each standard box at a profit of £1.

The company makes x luxury boxes and y standard boxes each hour.

The company wishes to maximise its profit, £P per hour.

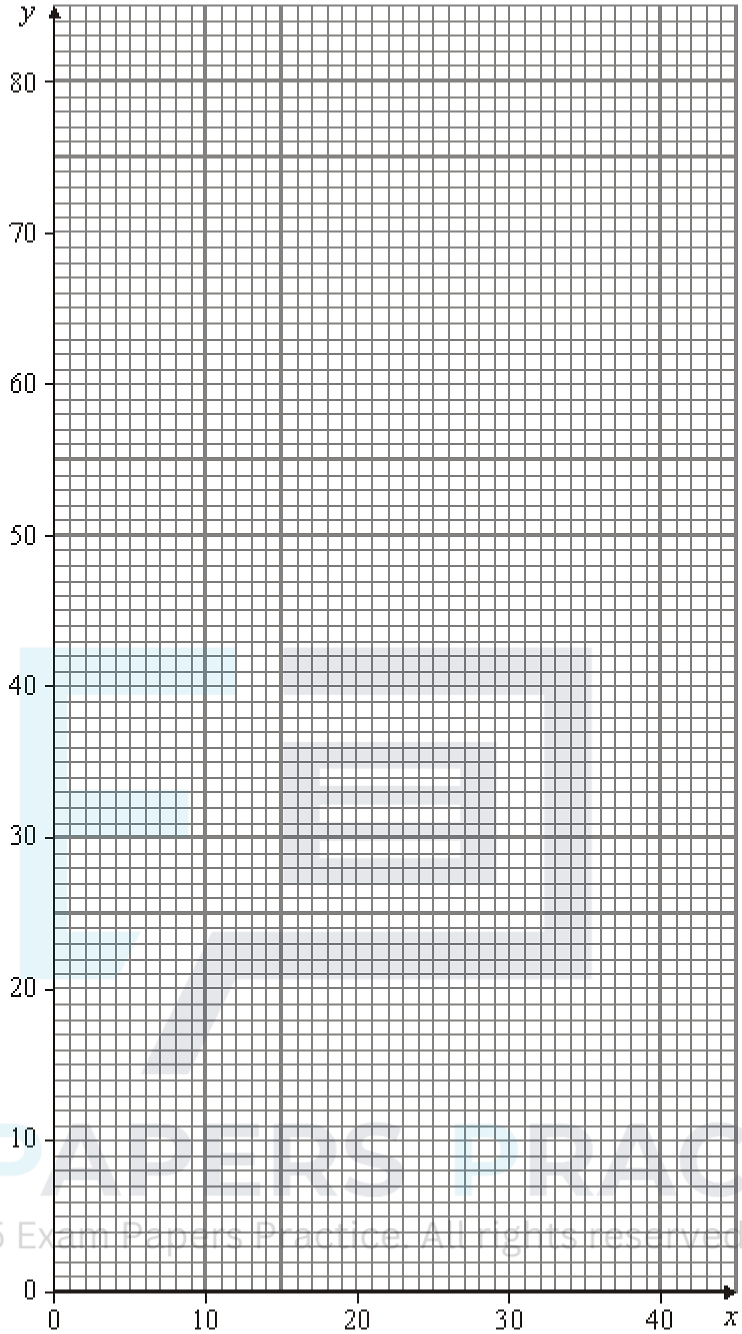
(a) Show that two of the constraints lead to the inequalities $2y \geq x$ and $2x + y \leq 60$.

(2)

(b) Formulate the company's situation as a linear programming problem.

(2)

(c) On the figure below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(5)

(d) Use your diagram to find the company's maximum hourly profit.

(2)

(e) The company changes its pricing structure so that the profit is now £2 for each box, for both luxury and standard. On a particular day the company makes boxes for exactly 7 hours. Only complete boxes are sold. Find the maximum profit on this day. State the two different combinations of boxes that correspond to this maximum.

(4)

(Total 15 marks)

Q24.

Every Saturday, David irons for the family. He irons three different types of clothing: dresses, shirts and jackets.

David knows that each Saturday:

he must iron at least 2 dresses, at least 5 shirts and at least 2 jackets; he can only iron at most 28 items of clothing; he takes 3 minutes to iron a dress, 2 minutes to iron a shirt and 2 minutes to iron a jacket. The iron must be used for at least 40 minutes but not more than 60 minutes.

Each Saturday, David irons x dresses, y shirts and z jackets.

(a) Find six inequalities that model David's situation.

(4)

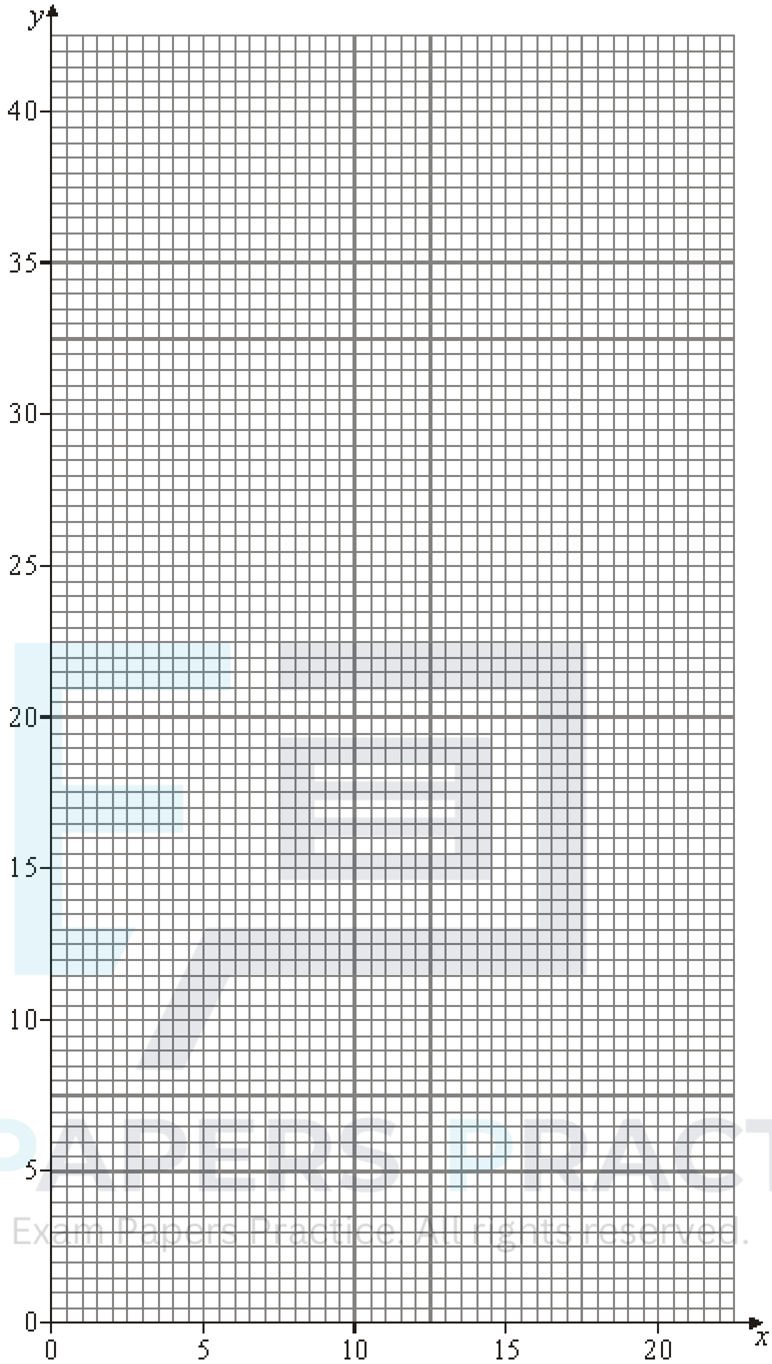
(b) On a particular Saturday, David irons the same number of jackets as dresses. David's aim is to iron the maximum number of items of clothing.

(i) Show that the inequalities found in part (a) simplify to

$$x \geq 2, \qquad y \geq 5, \qquad 2x + y \leq 28, \qquad 40 \leq 5x + 2y \leq 60.$$

(2)

(ii) On the diagram below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(7)

(iii) Use your diagram to find how many of each type of clothing David might iron to achieve his aim.

(2)

(Total 15 marks)

Q25.

Teresa is a florist. Every day she makes two types of bouquet, standard and luxury.

Each standard bouquet has 3 roses, 6 carnations and 4 lilies.

Each luxury bouquet has 6 roses, 3 carnations and 4 lilies.

Every day, Teresa has 600 roses, 600 carnations and 480 lilies available.

She makes a profit of £1.50 on each standard bouquet sold and £2.50 on each luxury bouquet sold. Each day, Teresa sells all the bouquets she makes.

Each day, Teresa makes x standard bouquets and y luxury bouquets, and she wishes to maximise her daily profit, £P.

(a) Show that x and y must satisfy the following inequalities.

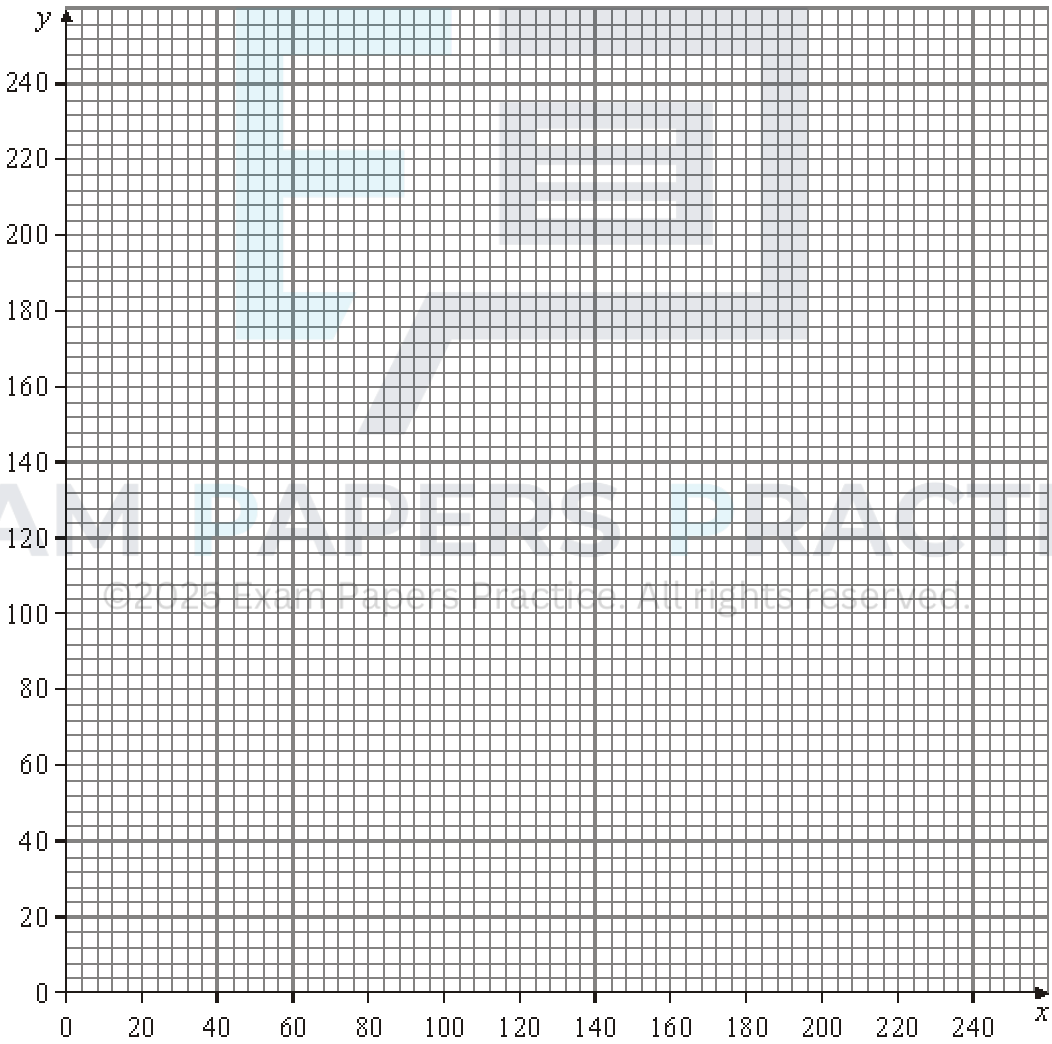
$$x + 2y \leq 200$$

$$2x + y \leq 200$$

$$x + y \leq 120$$

(3)

(b) Draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.



(7)

(c) Use your diagram to find Teresa's maximum daily profit.

(2)

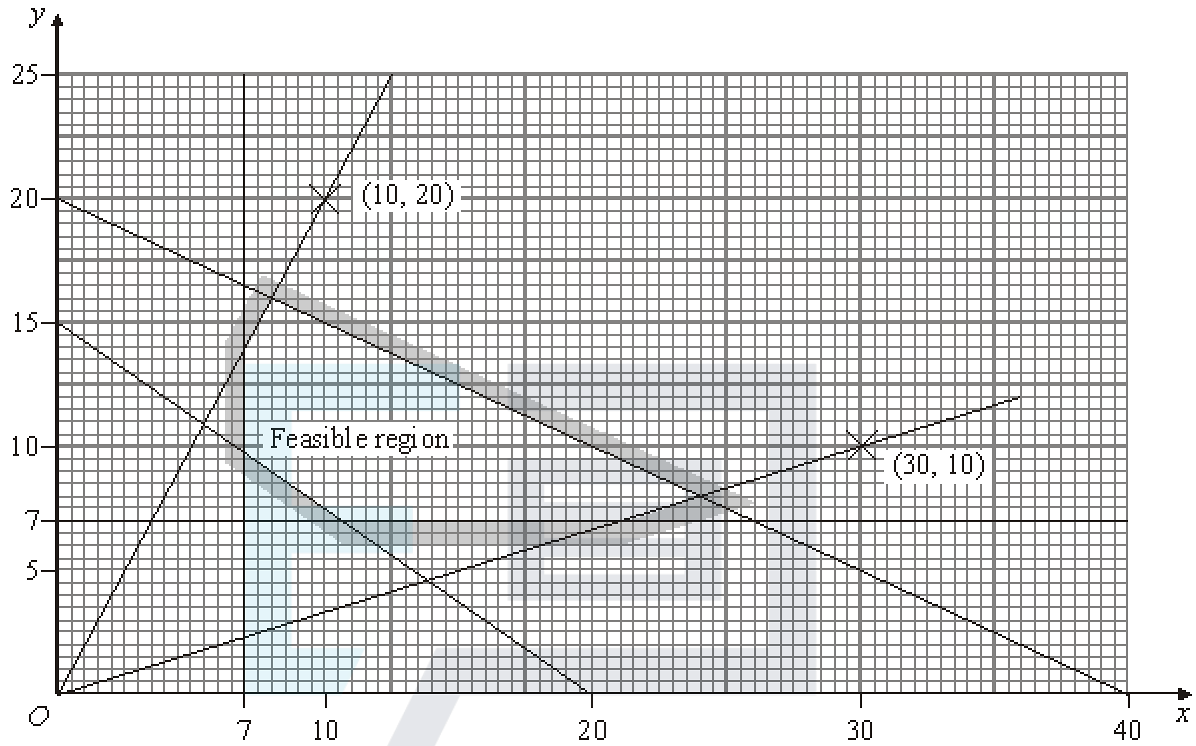
(d) One day Teresa decides to make more standard bouquets than luxury bouquets.

Find her maximum possible profit on this day.

(4)
(Total 16 marks)

Q26.

The following graph shows the feasible region of a linear programming problem.



(a) Find the **six** inequalities that define the feasible region.

(7)

(b) On the feasible region find the maximum value of $2x + 5y$ and state the coordinates of the point where this value occurs.

(3)
(Total 10 marks)

Q27.

A company produces two types of gift box, standard and luxury.

Each day, the company produces x standard and y luxury gift boxes.
Each day, the company must produce at least 20 of each type and at least 70 in total.

The boxes are produced using three different machines.

Machine A must be used for at least 100 minutes each day.
Machine B is available for a maximum of 5 hours each day.
Machine C is available for a maximum of 8 hours each day.

The time taken to produce a standard box is:

2 minutes on machine A;
3 minutes on machine B; and

4 minutes on machine C.

A luxury box requires:

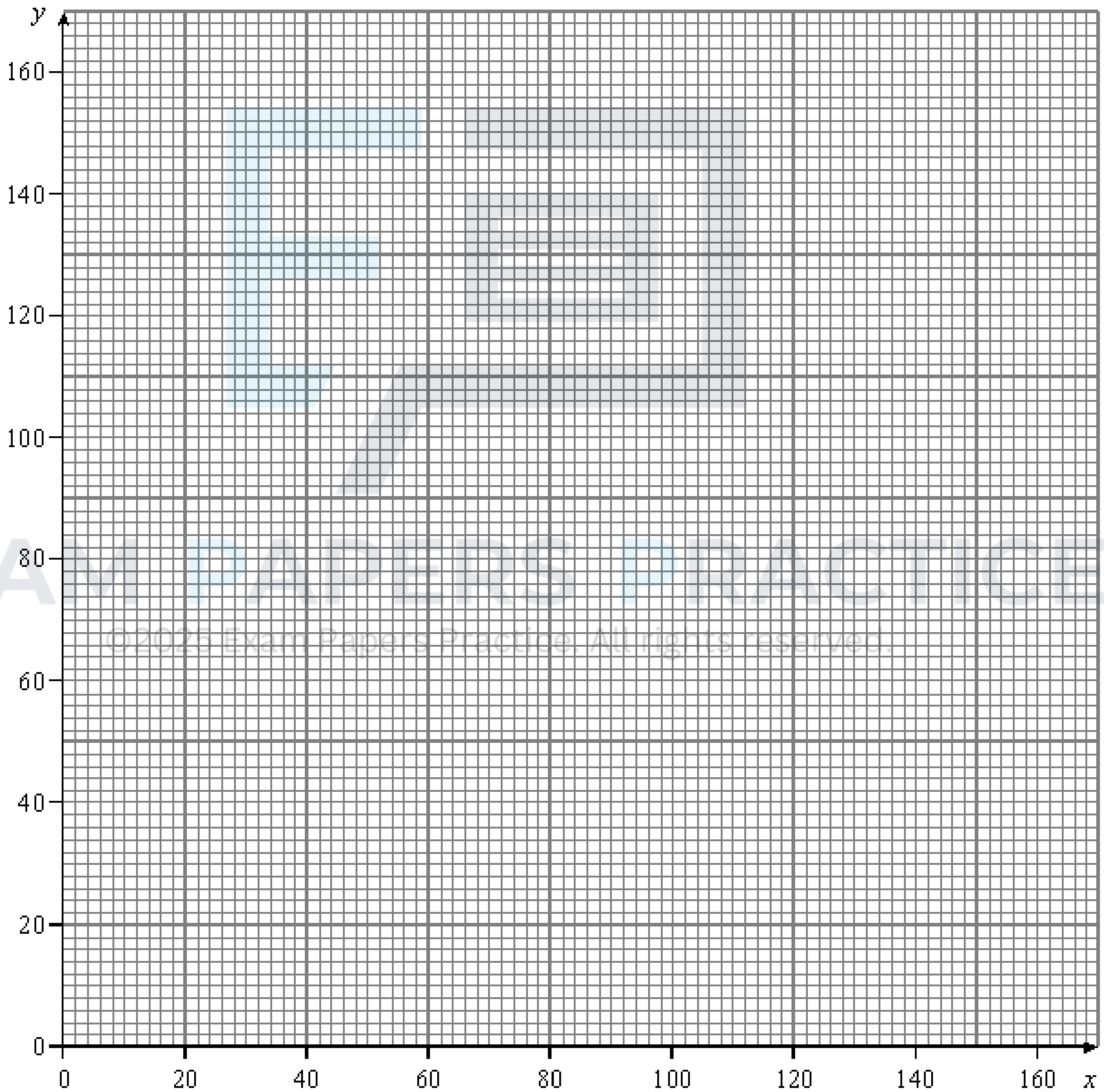
1 minute on machine A;
2 minutes on machine B; and
4 minutes on machine C.

The company wishes to produce the maximum number of boxes each day.

(a) Formulate the company's situation as a linear programming problem.

(6)

(b) On the diagram below, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region.



(6)

(c) (i) Find the maximum number of boxes the company can produce each day.

(2)

(ii) Find the number of different combinations of standard and luxury boxes that would enable the company to produce this maximum total.

(2)
(Total 16 marks)

Q28.

A football team is performing badly. The manager of the team can buy some new players to try and gain some extra points.

A very simple model of the manager's situation is as follows.

He can buy new attackers for £3 million each and new defenders for £1 million each.

Each attacker will help the club to get two more points and each defender will help the club to get one more point.

The manager has £12 million to spend.

The team must gain at least 5 more points.

The manager must buy at least one defender and at least one attacker.

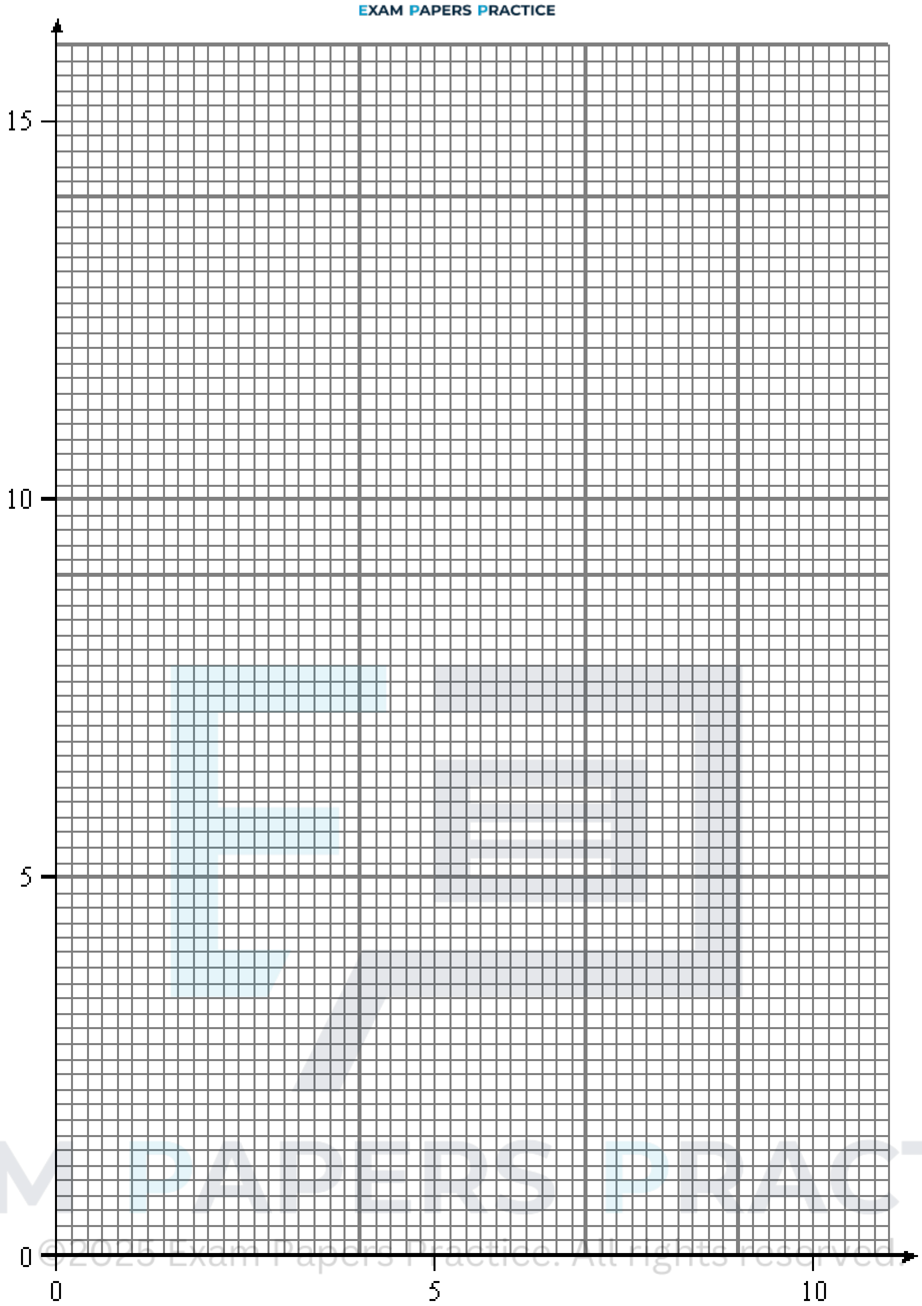
He must spend at least £6 million.

The manager buys x attackers and y defenders.

(a) Express the manager's situation in terms of inequalities suitable for investigating the problem by linear programming.

(4)

(b) On the graph below, draw a suitable diagram to enable the problem to be investigated graphically, indicating the feasible region.



(5)

(c) Use your diagram to find:

(i) the maximum number of points the team can gain;

(2)

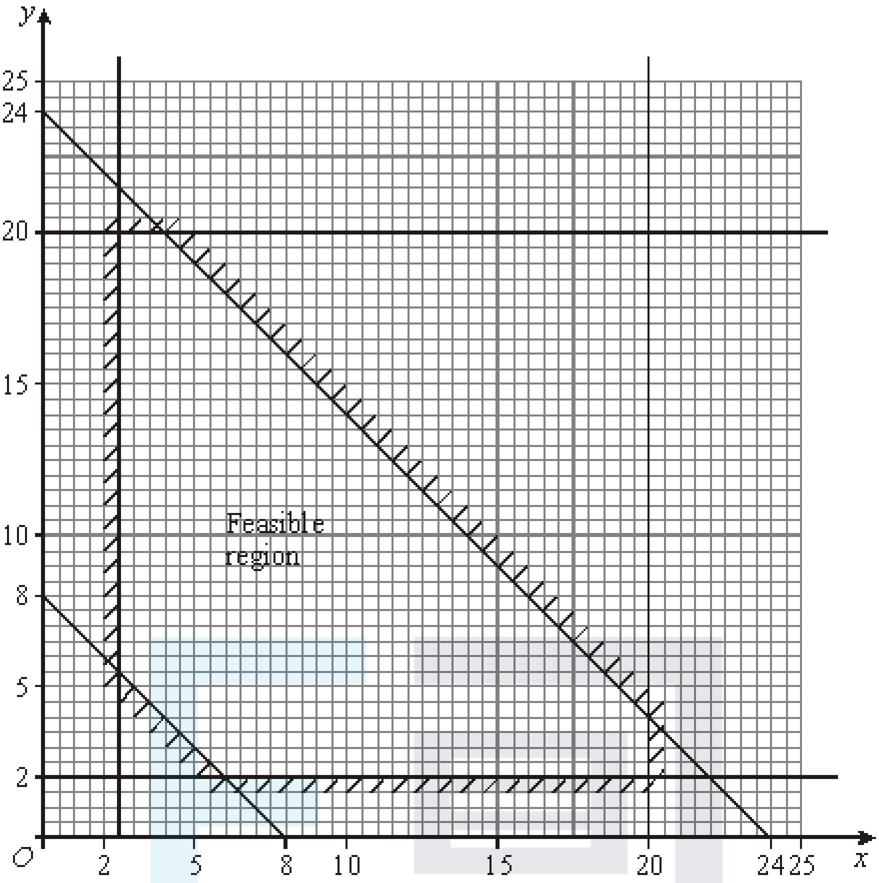
(ii) the number of attackers and defenders which will allow the team to gain the maximum number of points if the manager spends exactly £11 million.

(3)

(Total 14 marks)

Q29.

The following graph shows the feasible region of a linear programming problem.



(a) Find the six inequalities that define the feasible region.

(4)

(b) On this feasible region, find:

(i) the maximum value of $x + 3y$;

(2)

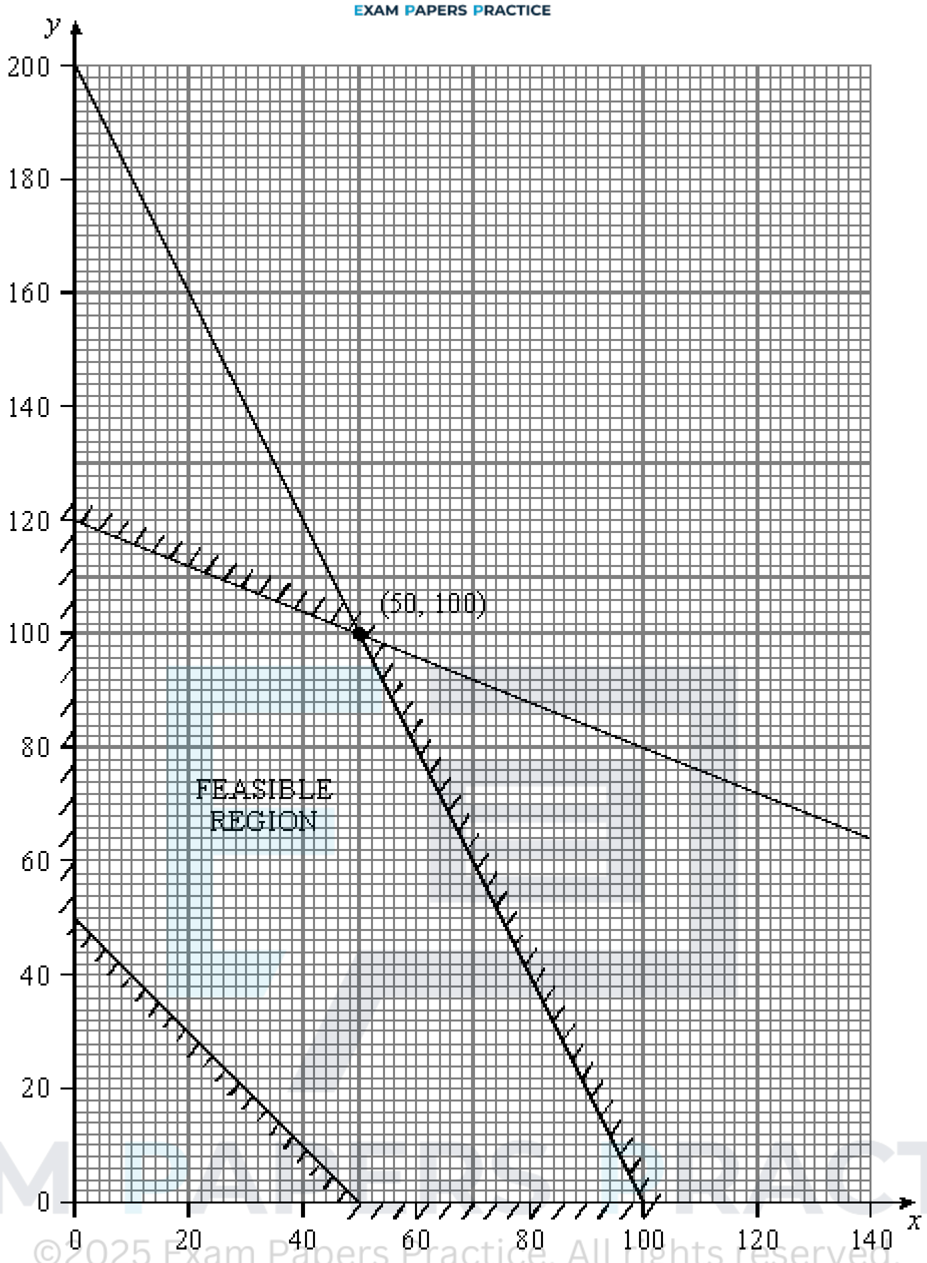
(ii) the minimum value of $-x + 2y$.

(3)

(Total 9 marks)

Q30.

The following graph shows the feasible region of a linear programming problem.



(a) On this feasible region find

(i) the maximum value of the function $2x + 3y$,

(3)

(ii) the minimum value of the function $4x + y$.

(2)

(b) Find the **five** inequalities that define the feasible region.

(6)

(Total 11 marks)

Q31.

The Tony television company makes analogue and digital televisions. Both types of television require a number of component A and component B.

Each analogue television requires 2 of component A and 3 of component B. Each digital television requires 4 of component A and 1 of component B.

Each day: the company has 50 of component A and 24 of component B available; and the company is to make at least 2 of each type of television, but no more than 20 in total.

The company sells each analogue television at a profit of £20 and each digital television at a profit of £25. Each day, the company makes and sells x analogue and y digital televisions. The company needs to find its minimum and maximum daily income, £T.

(a) Formulate the company's situation as a linear programming problem.

(5)

(b) On **Figure 1**, draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.

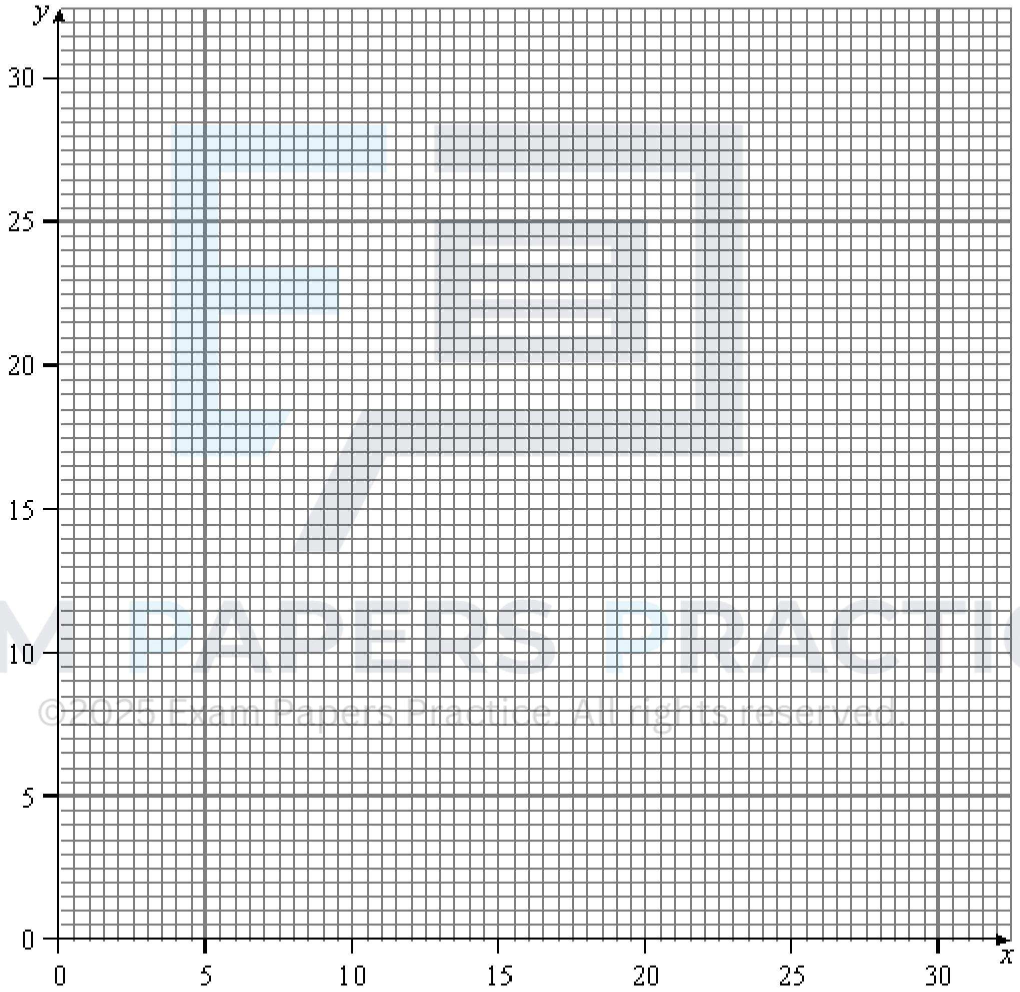


Figure 1

(6)

(c) Use your diagram to find the company's minimum and maximum daily income, £T.

(6)

(Total 17 marks)

Q32.

A linear programming problem is represented by the following:

$x + y \geq 20;$

$y \leq 2x;$

$x + y \leq 60;$

$x + 2y \leq 80;$

maximise $F = 2x + 5y$.

(a) On **Figure 1,** draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.

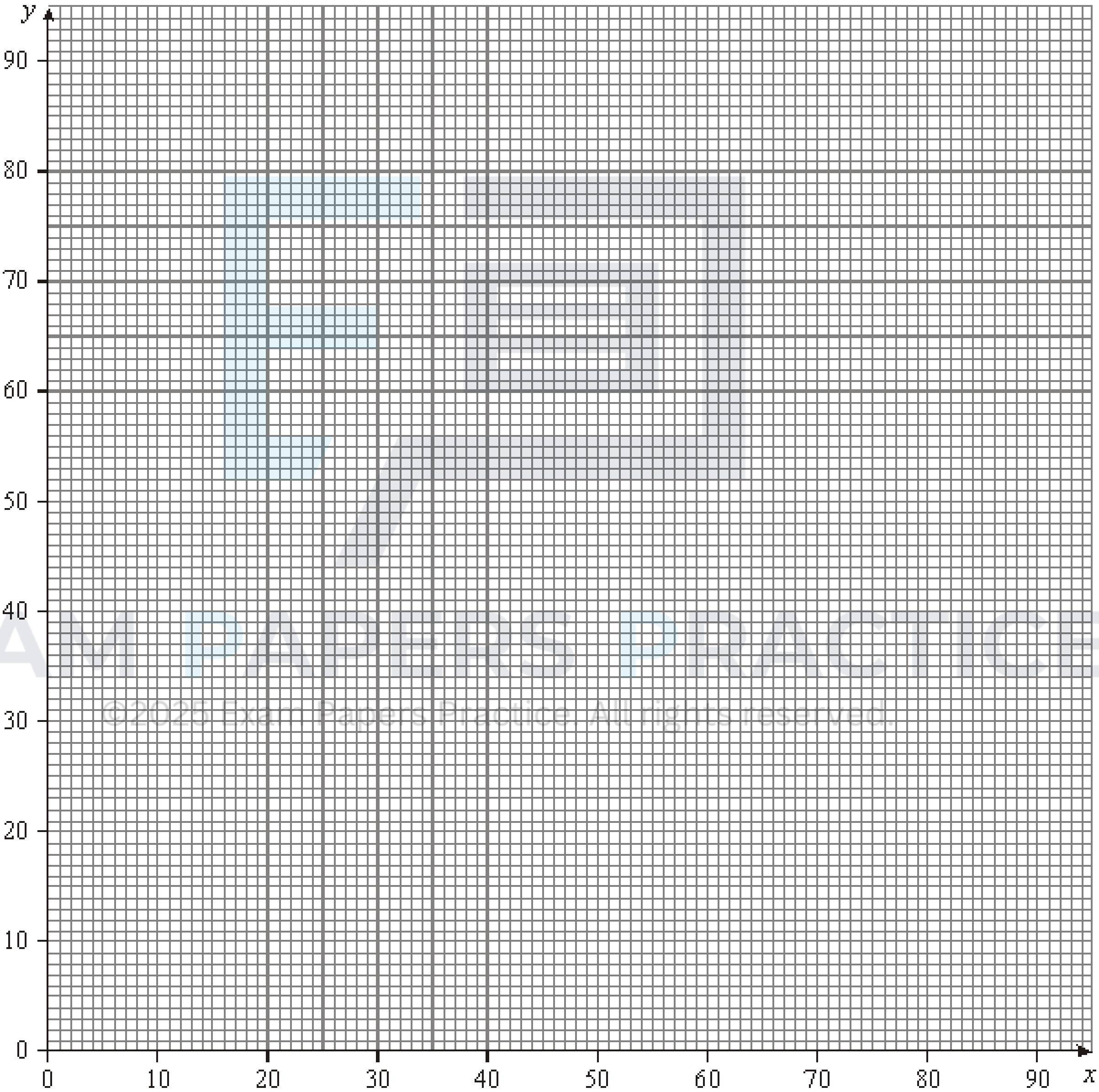


Figure 1

(7)

(b) Use your diagram to find the maximum value of F on the feasible region.

(2)

(Total 9 marks)

Q33.

[Graph paper is provided for use in answering this question.]

The Elves toy company makes toy trains and dolls' prams, which use the same wheels and logo stickers.

Each train requires 8 wheels and 2 logo stickers. Each pram requires 8 wheels and 3 logo stickers.
The company has 7200 wheels and 2200 logo stickers available.
The company is to make at least 300 of each type of toy and at least 800 toys in total.
The company sells each train for £20 and each pram for £25.
The company makes and sells x trains and y prams.
The company needs to find its minimum and maximum total income, £T.

(a) Formulate the company's situation as a linear programming problem.

(5)

(b) Draw a suitable diagram to enable the problem to be solved graphically, indicating the feasible region and the direction of the objective line.

(6)

(c) Use your diagram to find the company's minimum and maximum total income, £T.

(4)

(Total 15 marks)

Q34.

A garage owner is to restock with new cars of three different types; hatchbacks, sports cars and estate cars.

She decides that:

the number of hatchbacks must be greater than the combined total number of sports cars and estates;
the number of sports cars must not exceed 15% of the total number of cars;
there must be at least as many sports cars as estates;
the number of estates must be less than 60% of the number of hatchbacks.

Each hatchback costs £8 000, each sports car costs £12 000, and each estate costs £14 000.

The garage owner buys x hatchbacks, y sports cars and z estates.

She wishes to minimise her total cost, £C.

(a) Formulate the above situation as a linear programming problem, simplifying your inequalities so that all the coefficients are integers.

(7)

(b) The garage owner buys 6 sports cars. By using a graphical method, find her minimum total cost in this case.

(6)
(Total 13 marks)

Q35.

A building company has acquired a building site on which to build residential dwellings. It has permission to build at least 12 but not more than 16 dwellings. The company can build a combination of houses and bungalows on the site but it must build at least 4 of each.

Each house needs a plot of size 450 m^2 and will have a floor area of 200 m^2.
Each bungalow needs a plot of size 600 m^2 and will have a floor area of 150 m^2.
The cost of a plot is £50 per m^2 and the building costs are £100 per m^2 of floor area. The total value of land which can be used for building plots must not exceed £450 000. The total building costs must not exceed £300 000.

The company builds x houses and y bungalows.

(a) Show that the company's situation can be modelled by the following inequalities.

$12 \le x + y \le 16 \qquad x \ge 4 \qquad y \ge 4 \qquad 3x + 4y \le 60 \qquad 4x + 3y \le 60$

(5)

(b) Draw a suitable diagram to represent this problem graphically, indicating the feasible region.

(5)

(c) The company sells all the houses and bungalows at a profit of £10 000 per dwelling. Find the **minimum** profit that the company is sure to make on this building site.

List **all** the different pairs of values of x and y that would produce this minimum profit.

(2)

(d) An alternative pricing structure is proposed in which houses and bungalows are all sold at the same **price** of £S. The company **maximises** the profit and this total profit is £190 000.

Find the largest and smallest possible values of S.

(6)

(Total 18 marks)

Q36.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

A knitwear company employs machinists to make cardigans and jumpers on knitting machines.

In a week a machinist makes x cardigans and y jumpers.

Each machinist must make at least 5 cardigans and at least 20 jumpers each week.

Each cardigan uses 3 ounces of wool and has 6 buttons.

Each jumper uses 3.5 ounces of wool and has 2 buttons.

In any one week a machinist has only 210 ounces of wool and 180 buttons available.

A machine can knit a cardigan in 30 minutes and a jumper in 40 minutes. To ensure efficiency, the company requires each machinist to use a machine for at least 20 hours each week.

(a) Find **five** inequalities that model the machinist's situation.

(4)

(b) Draw a suitable diagram to represent this problem graphically, indicating the feasible region.

(5)

(c) The machinist is paid £3.00 for each cardigan and £2.50 for each jumper that he makes.

(i) Draw an objective line that will represent his pay, £P, for the week.

(2)

(ii) Find values of x and y that will maximise his weekly pay and calculate this pay.

(2)

(d) The pay structure is to be altered so that he will be paid £2.50 for each cardigan and £4.00 for each jumper. Determine his new maximum pay.

(4)
(Total 17 marks)

Q37.

[One sheet of 2mm graph paper is provided for use in answering this question.]

A company is making two types of door, standard and luxury. Both types of door require the use of two different machines A and B.

Both types of door require 90 minutes on machine A. A standard door requires 60 minutes on machine B but a luxury door requires 120 minutes on this machine.
During any one week machine A can be used for a maximum of 20 hours and machine B can be used for a maximum of 25 hours.

The company makes a profit of £10 on each standard door and £12 on each luxury door. In a week the company makes x standard doors and y luxury doors.

(a) Show that the above information can be modelled by the following inequalities

$$x \geq 0, \qquad y \geq 0, \qquad 3x + 3y \leq 40 \qquad x + 2y \leq 25$$

(3)

(b) (i) Draw a suitable diagram to represent the problem graphically, indicating the feasible region.

(5)

(ii) Draw an objective line that will represent the company's profit for the week, £P. Hence indicate the vertex, V, that could correspond to the maximum value of P.

(3)

(iii) State why this maximum value of P cannot be achieved

(1)

(iv) Find values of x and y that will maximise the company's profit and calculate this profit.

(3)
(Total 15 marks)

Q38.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

A factory manufactures three items; screws, nuts and bolts.

The items are first produced on a machine which is available for 4 hours per day. The machine takes 6 minutes to produce a screw, 4 minutes to produce a nut and 2 minutes to produce a bolt. The items are then cleaned on a machine that is available for 55 minutes per day. Each screw takes 30 seconds to clean, each nut takes 40 seconds to clean and each bolt takes 60 seconds to clean.

(a) An apprentice at the factory produces x screws, y nuts and z bolts in a day. Find and simplify **two** inequalities each involving x, y and z, that model the conditions given above.

(3)

(b) As a further requirement the apprentice must produce the same number of bolts as nuts each day.

(i) Show that, with this additional constraint, the two inequalities found in part (a) become

$$x + y \leq 40 \qquad \text{and} \qquad 3x + 10y \leq 330.$$

(2)

(ii) Draw a suitable diagram to represent this problem graphically, indicating the feasible region.

(4)

(iii) The apprentice has to make the largest possible total number of items each day. Draw an objective line that will represent his total daily output, T. Hence indicate the vertex that will correspond to the maximum value of T.

(3)

(iv) Calculate the maximum value of T.

(1)

(v) On a particular day the cleaning machine is available for only 48 minutes. Find the maximum number of items that the apprentice can make during this particular day.

(4)

(Total 17 marks)

Q39.

[One sheet of 2 mm graph paper is provided for use in answering this question.]

An insurance saleswoman can sell two types of policies, pension policies and life assurance policies.

Head-office require her to sell at least three policies of each type per week. Regulations stipulate that a pension policy needs 60 minutes of explanation but a life assurance policy needs 20 minutes of explanation.

The saleswoman knows that she can sell policies only in an evening and that in a normal week she has 15 hours available to sell policies.

Each pension policy has an annual premium income (API) of £600 whereas each life assurance policy has an API of £120.

Head-office requires a sales person to sell, in a week, policies with a total API of at least

£4800.

The saleswoman is paid a commission on each policy she sells. For each pension policy she is paid £48 and for each life assurance policy she is paid £36.

In a week the saleswoman sells x pension policies and y life assurance policies.

(a) (i) Show that $3x + y \leq 45$.

(1)

(ii) Find **three** further inequalities in x and y that model the saleswoman's situation.

(3)

(b) (i) Draw a suitable diagram to represent this problem graphically, indicating the feasible region.

(5)

(ii) Draw the line that will represent a weekly commission of £576. Hence find the vertex that will maximise her weekly commission.

(2)

(c) (i) State the values of x and y that maximise her commission.

(2)

(ii) If the saleswoman is paid this commission each week throughout a 45 week working year, calculate her income for the year.

(1)

(d) During a promotional week head-office changes her commission structure so that a pension policy pays £90 a life assurance policy pays £18. All other constraints remain the same.

(i) She was paid £1170 in commission for the week. Determine **one** possible pair of values of x and y.

(2)

(ii) Calculate her maximum possible commission for the week.

(3)

(Total 19 marks)

Q40.

[One sheet of 2mm graph paper is provided for use in answering this question.]

A businessman is considering starting a car hire company using Alpha cars which cost £12500 each and Beta cars which cost £7500 each.

Research suggests that demand is such that he will need at least ten cars, but he has only £187 500 to buy cars.

Annual insurance and service costs are estimated at £1000 for each Alpha car and £2000 for each Beta car, and he wishes to limit his total annual expenditure on these items to £29000.

Assume that he buys x Alpha and y Beta cars.

(a) Show that the above information can be modelled by the following inequalities.

$$x + y \geq 10 \qquad 5x + 3y \leq 75 \qquad x + 2y \leq 29$$

(3)

(b) The businessman estimates that he will make an annual profit of £1200 per car on both Alpha and Beta cars.

(i) Draw a suitable diagram to represent this problem graphically, indicating the feasible region.

(5)

(ii) Draw an objective line that will represent his annual profit, £P. Hence indicate the vertex that will correspond to the maximum value of P.

(3)

(iii) Write down the values of x and y which maximise P, and calculate the maximum value.

(3)

(Total 14 marks)

Q41.

A jeweller works for up to seven days a week. He makes necklaces and bracelets. He makes one necklace on each working day and on some working days he also makes a bracelet. He makes a total of at most 10 items each week.

Assume that the jeweller makes x necklaces and y bracelets in a week.

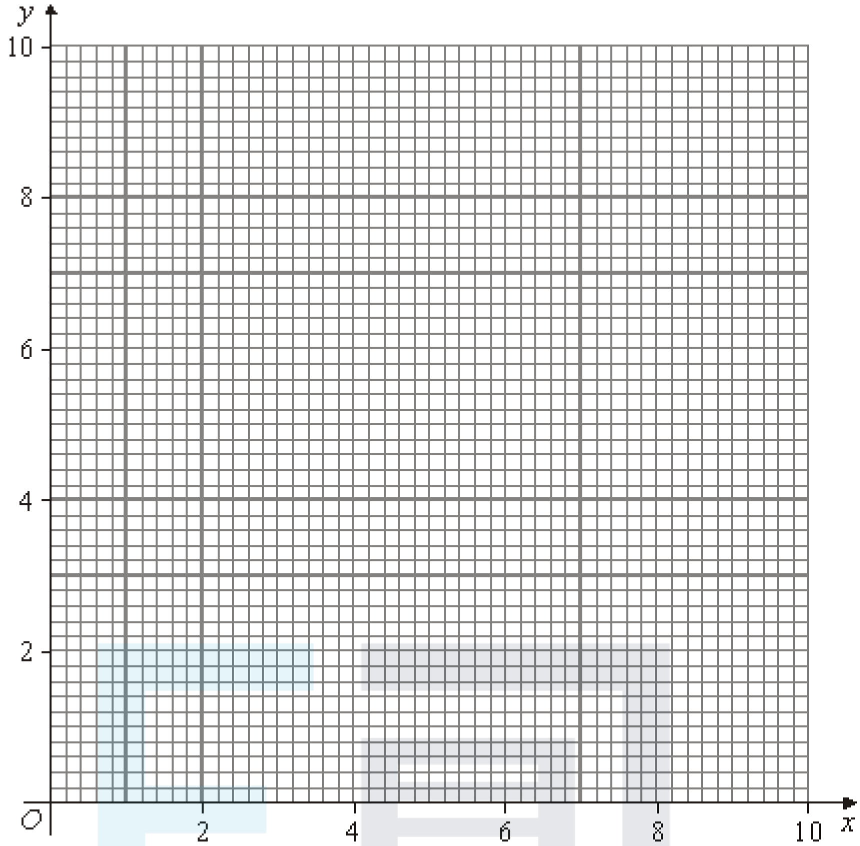
(a) Explain why x and y must satisfy

$$0 \leq x \leq 7, \quad 0 \leq y \leq 7 \text{ and } \quad y \leq x$$

and write down another inequality which x and y must satisfy.

(3)

(b) On the figure below illustrate the region representing those (x, y) which satisfy all the inequalities in part (a).



(4)

(c) The jeweller charges £20 for a necklace and £10 for a bracelet.

(i) Find how many necklaces and bracelets he should make each week in order to maximise his income.

(4)

(ii) The jeweller wants to increase the cost of the bracelets but leave the cost of the necklaces unchanged. He wants to choose the new price so that when maximising his income he need only work five days a week. Calculate the minimum that he must charge for each bracelet.

(4)
(Total 15 marks)

Q42.

Pisa Pizzas make two types of pizza base, the Romano and the Sardino.

The Romano requires 150 grams of flour, 50 grams of butter and 1 egg.
The Sardino requires 300 grams of flour, 20 grams of butter and 1 egg.

The available ingredients are $7\frac{1}{2}$ kg of flour, 1 kg of butter and 30 eggs.

Assume that Pisa Pizzas make x Romanos and y Sardinos.

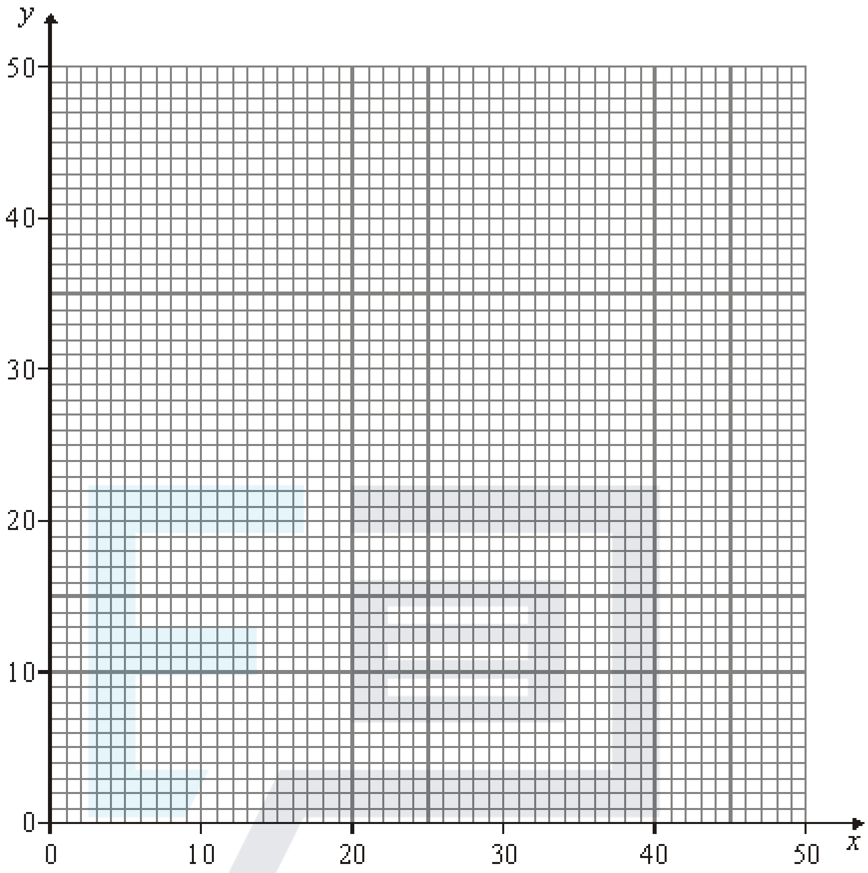
(a) Show that the flour and butter requirements imply that x and y must satisfy

$$x + 2y \leq 50 \quad \text{and} \quad 5x + 2y \leq 100$$

and deduce an inequality from the egg requirements.

(3)

(b) On the figure below, illustrate the region of those (x, y) which satisfy $x \geq 0$, $y \geq 0$ and the three inequalities in part (a).



(4)

(c) Pisa Pizzas make £1 profit on each Romano and £1.50 profit on each Sardino.

Find how many of each type they should make in order to maximise their profit.

(4)

(d) If instead Pisa Pizzas made £2 profit on each Romano and £1.50 profit on each Sardino, how many of each type should they make in order to maximise their profit?

(4)

(Total 15 marks)

Q43.

At Basehome Do-it-yourself Store, nails, screws and hooks can be bought in two types of mixed pack, the Xperience pack and the You-do-it pack.

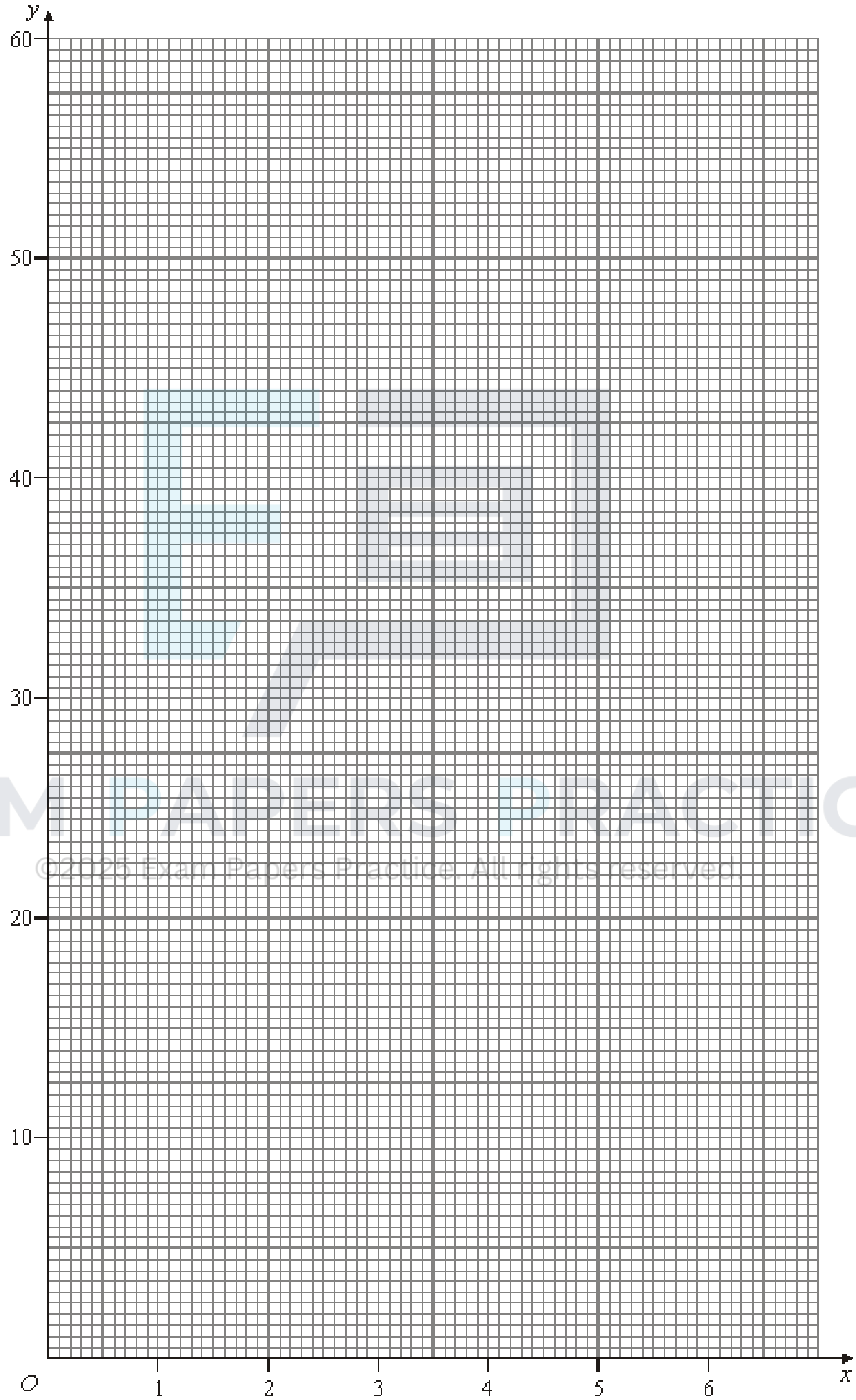
The Xperience pack contains 1000 nails, 1500 screws and 500 hooks. It costs £6. The You-do-it pack contains 100 nails, 100 screws and 100 hooks. It costs £1. A housebuilder needs 5000 nails, 6000 screws and 3000 hooks.

Assume that the housebuilder buys x Xperience packs and y You-do-it packs.

(a) Show that the inequality $10x + y \geq 50$ ensures that the housebuilder has enough nails and write down the corresponding inequalities for the screws and the hooks.

(4)

(b) On the graph below, illustrate the region of those x and y which satisfy $x \geq 0$, $y \geq 0$ and the three inequalities from part (a).



(5)

(c) Calculate how many packs of each type the housebuilder should buy in order to satisfy his requirements at the minimum cost.

(4)

(d) The Basehome Do-it-yourself Store has a “3-for-the-price-of-2” offer: if you buy 3 identical items you only have to pay for 2 of them. If the housebuilder takes advantage of this offer, how many packs of each type should he buy in order to satisfy his requirements at minimum cost?

(3)

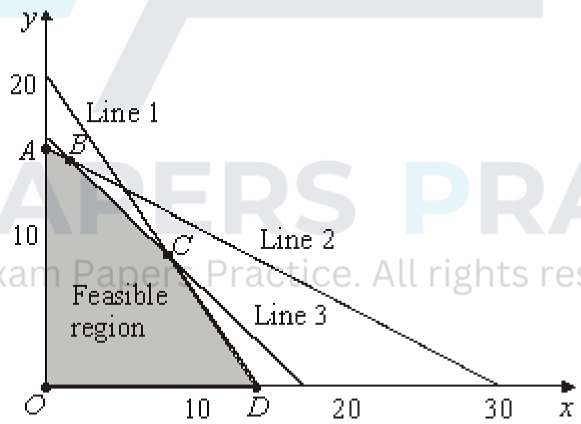
(Total 16 marks)

Q44.

A factory can produce two types of tractor, the Xtreme and the Yltra. The factory makes x of the Xtremes and y of the Yltras. In order to maximise the profit, P, the following linear programming problem has to be solved:

$$\text{Maximise } P = 2x + 3y$$
$$\text{subject to } x \geq 0,\ y \geq 0$$
$$x + 2y \leq 30$$
$$3x + 2y \leq 42$$
$$x + y \leq 16$$

The diagram below is a sketch in which the feasible region for this problem is shaded.



(a) State the equations of Lines 1, 2 and 3.

(2)

(b) Solve the simultaneous equations

$$3x + 2y = 42$$
$$x + y = 16$$

and hence state the coordinates of the vertex C.

(3)

(c) The coordinates of the other four vertices of the feasible region are

O (0, 0) A (0, 15) B (2, 14) D (14, 0).

Solve the above linear programming problem completely.

(4)

(d) The factory finds that it has an additional constraint that at most 20% of the tractors which it makes can be Yltras.

(i) Show that this extra condition is equivalent to $4y \le x$.

(2)

(ii) Calculate how many of each type should now be made in order to maximise P.

(4)

(Total 15 marks)

Q45.

An athlete wants to supplement his diet with vitamins. He can buy either Xtravim tablets or Yeasty tablets.

Each Xtravim tablet contains 3 mg of vitamin A, 1 mg of vitamin B and 1 mg of vitamin C.

Each Yeasty tablet contains 1 mg of vitamin A, 4 mg of vitamin B and 1 mg of vitamin C.

The athlete's minimum daily requirement is 15 mg of vitamin A, 20 mg of vitamin B and 11 mg of vitamin C.

Assume that the athlete takes x Xtravim tablets and y Yeasty tablets each day.

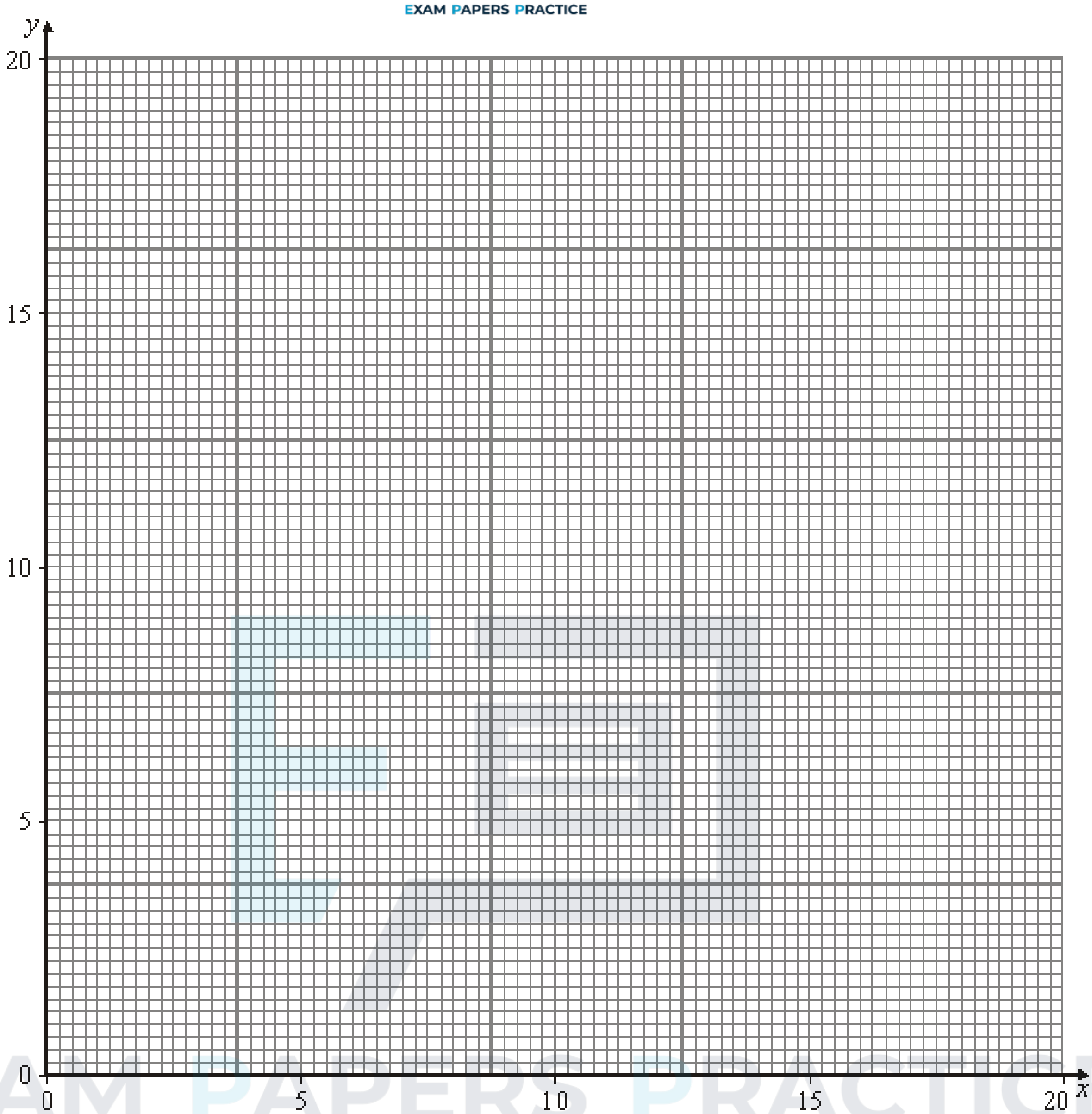
(a) Show that, in order to satisfy the athlete's requirement of vitamin A, x and y must satisfy

$$3x + y \ge 15$$

and write down the corresponding inequalities for the vitamin B and the vitamin C.

(3)

(b) On the figure below, illustrate the region of those (x, y) which satisfy $x \ge 0$, $y \ge 0$ and the three inequalities in part (a).



(4)

(c) The Xtravim tablets cost 2p each and the Yeasty tablets cost 5p each.

(i) Find how many tablets of each type the athlete should take each day in order to satisfy his vitamin requirements at minimum cost.

(4)

(ii) The Xtravim tablets are also available more cheaply by mail order. The new minimum cost to the athlete can still be achieved by continuing to take the same mixture of tablets as before or by taking a sufficient number of Xtravim tablets alone. Find the cost of Xtravim tablets when bought by mail order.

(4)

(Total 15 marks)

Q46.

[Graph paper should be provided for answering this question.]

The Carryit Company has 1000 large packing cases to transport from Liverpool to

Southampton; x of them by road, y of them by rail, and the rest by sea. The company intends to transport at least 10% and at most 50% of the cases by sea. Furthermore, the number transported by sea must be at most double the number transported by road.

(a) Write down, in terms of x and y, the number of cases transported by sea. Hence show that x and y must satisfy the inequalities:

$$500 \le x + y \le 900 \quad \text{and} \quad 3x + y \ge 1000$$

(3)

(b) Illustrate on graph paper the region of those (x, y) which satisfy $x \ge 0$ and $y \ge 0$ together with the inequalities in part (a).

(5)

(c) The cost of transporting the cases is

£50 for each case by road;
£55 for each case by rail;
£25 for each case by sea.

(i) Show that the total cost £T of sending the 1000 cases is given by

$$T = 25x + 30y + 25\,000$$

(1)

(ii) Hence use part (b) to find the minimum cost of transporting all the cases. State how many of the cases should be sent by each method in order to achieve that minimum.

(4)

(iii) In order to be more competitive the rail operator wants to reduce the cost of transporting each case to £R, where R is a whole number. The costs of transporting by road and sea will remain the same.

The rail operator chooses R as large as possible so that when minimising its transport costs the Carryit Company will have to send most of the 1000 cases by rail. Calculate the value of R.

(5)

(Total 18 marks)

Q47.

[A sheet of graph paper should be provided for answering part (d) of this question.]

The Bumper Hamper Company offers a selection of hampers containing tins of chicken, pots of caviar and jars of peaches.

Each tin of chicken weighs 300 grams, each pot of caviar weighs 100 grams and each jar of peaches weighs 200 grams.

Each hamper contains at most 11 items and weighs at most 2200 grams.

The company makes £4 profit on each tin of chicken, £3 profit on each pot of caviar and £1 profit on each jar of peaches.

Let a hamper contain x tins of chicken, y pots of caviar and z jars of peaches.

(a) Express each of the **two** restrictions on a hamper's contents as an inequality in x, y and z.

(2)

(b) Write down the total profit £P on a hamper in terms of x, y and z.

(1)

(c) The company decides to include just one jar of peaches in each hamper, but then to choose the amount of chicken and caviar in order to maximise its profits. Show that this is equivalent to maximising

$$P = 4x + 3y + 1$$

subject to

$$x \geq 0, \quad y \geq 0, \quad x + y \leq 10$$

and **one** other inequality, which you should state.

(3)

(d) Use a graphical method to solve the linear programming problem in part (c).

(8)

(e) The company drops the restriction of having to include peaches in each hamper. Find the maximum profit it can then make on a hamper.

(4)

(Total 18 marks)

Q48.

A baker wishes to make a batch of loaves to sell at the local market. He can make two types of loaf: the "cottage" and the "farmhouse".

Each cottage loaf requires 500 grams of flour and 50 grams of butter.
Each farmhouse loaf requires 300 grams of flour and 150 grams of butter.
The baker has available 4.5 kg of flour and 1.5 kg of butter.
The baker can make at most 12 loaves.

Assume that the baker produces x cottage loaves and y farmhouse loaves.

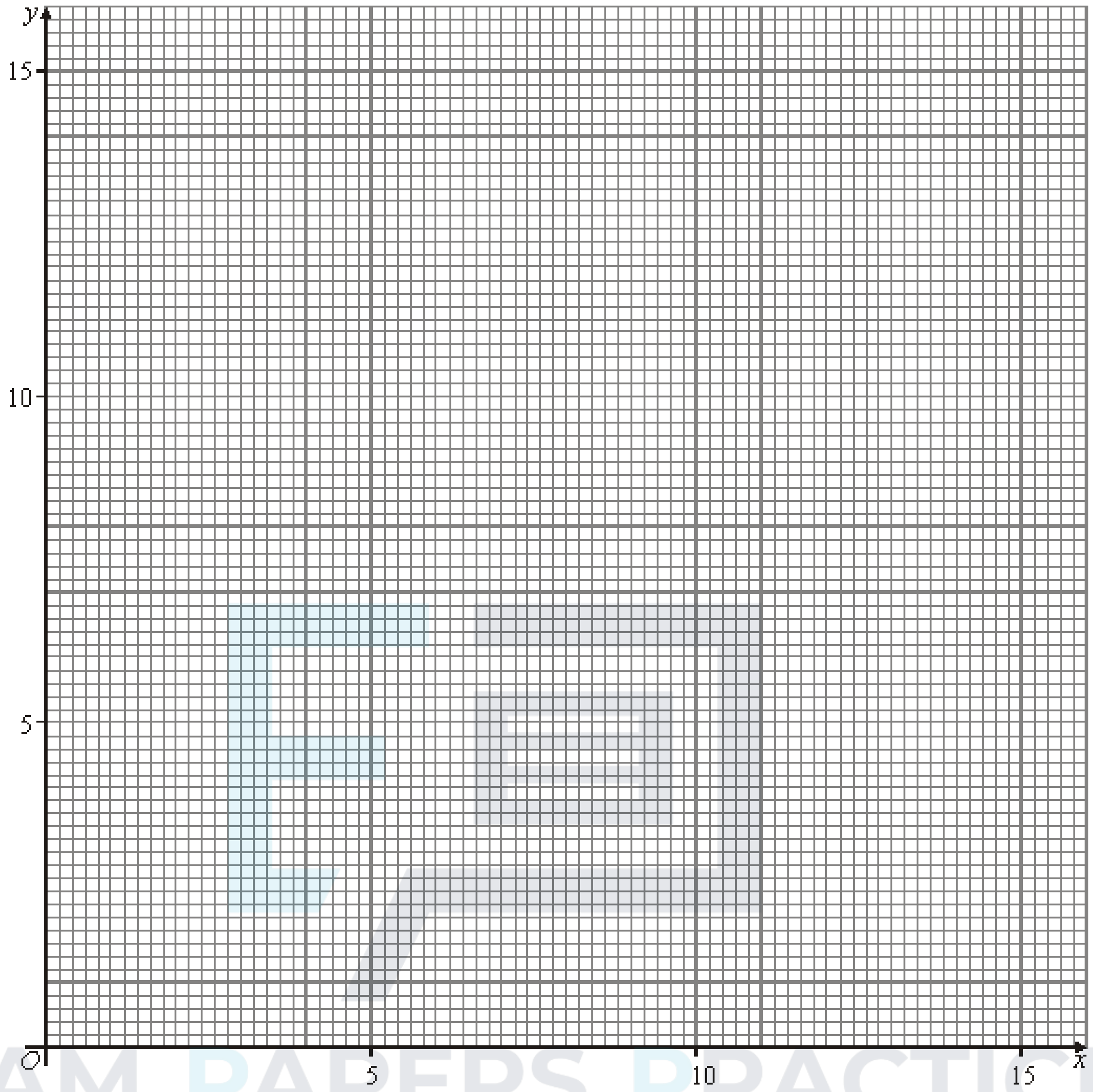
(a) (i) Show that the flour requirements imply that $5x + 3y \leq 45$ and write down the corresponding inequality for the butter requirements.

(3)

(ii) Write down an inequality which is equivalent to the fact that the baker can make at most 12 loaves.

(1)

(b) On the graph paper below illustrate the region of those x and y which satisfy $x \geq 0$, $y \geq 0$ and the three inequalities from part (a)



(4)

(c) The baker makes 30p profit on each cottage loaf and 60p profit on each farmhouse loaf. Calculate how many of each type he should make in order to maximise his profit.

(4)

(d) The baker decides to put up the price of the cottage loaves. This means that he will make 90p profit on each cottage loaf and 60p on each farmhouse loaf. Calculate the maximum profit which he will be able to make.

(4)

(Total 16 marks)

Q49.

A class decides to build a model of its school using toy bricks.

The model will require 6000 red bricks, 9000 blue bricks and 12 000 green bricks. They can buy the bricks in mixed boxes, the Xtra Box and the Ybuild Box.

Each Xtra Box contains 100 red bricks, 200 blue bricks and 300 green bricks.

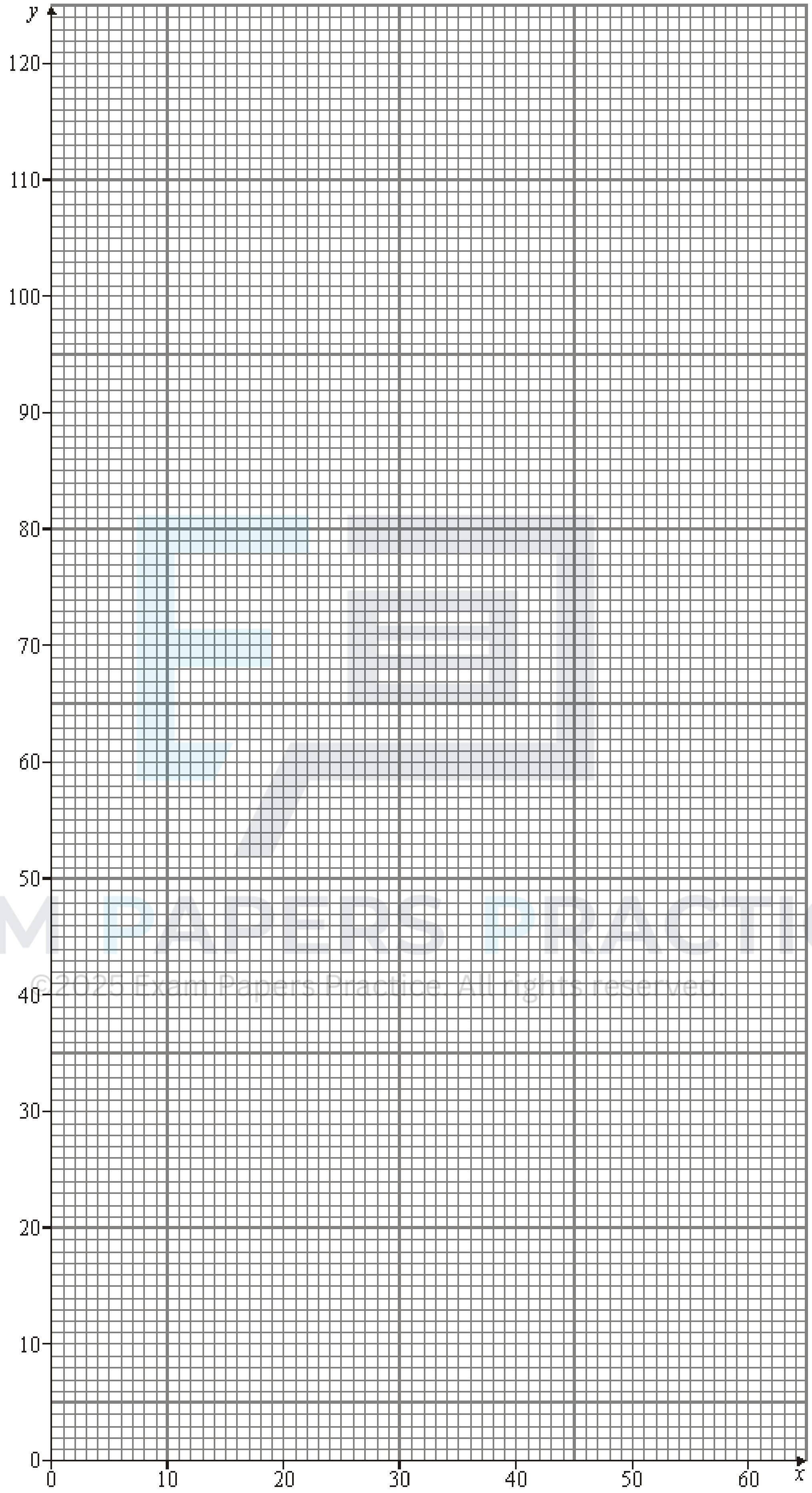
Each Ybuild Box contains 200 red bricks, 100 blue bricks and 100 green bricks.

(a) Let x be the number of Xtra Boxes and y the number of Ybuild Boxes bought by the class. Write down three inequalities in x and y which will ensure that the class has enough red, blue and green bricks.

(3)

(b) On the graph paper on the next page illustrate the region of those (x, y) with $x \geq 0$, $y \geq 0$ which satisfy your three inequalities from part (a).

(4)


y
120
110
100
90
80
70
60
50
40
30
20
10
0
0
10
20
30
40
50
60
x

(c) What is the minimum total number of boxes which the class must buy?

(4)

(d) The class then changes its plans which means that the model will require more red bricks than originally expected. They find that the minimum number of boxes now needed to satisfy all the requirements is 55. Find how many of those boxes will be Xtra Boxes and how many will be Ybuild Boxes.

(3)

(Total 14 marks)

Q50.

A tropical fish collector wishes to fill an aquarium with two types of fish, X-ray fish and Yellowtails. He requires that:

there are at least two X-ray fish;
there are at least as many Yellowtails as X-ray fish;
there are at least 6 fish and at most 10 altogether.

(a) Let x be the number of X-ray fish and y the number of Yellowtails. Write down inequalities x and in y which represent the above conditions.

(3)

(b) On graph paper illustrate the region of those x and y which satisfy the inequalities in part (a).

(6)

(c) The cost of each X-ray fish is £2 and the cost of each Yellowtail is £4. Find the minimum and maximum total costs of a collection of fish satisfying the above conditions

(4)

(d) List all the possible total costs of a collection of fish satisfying the above conditions.

(2)

(Total 15 marks)

Q51.

A strawberry grower decides to make jars of jam and jars of jelly for sale. The required ingredients for the two recipes are

Jam recipe (makes 2 jars)	**Jelly recipe (makes 10 jars)**
0.5 kg strawberries	6 kg strawberries
1 kg sugar	7 kg sugar
1 lemon	2 lemons

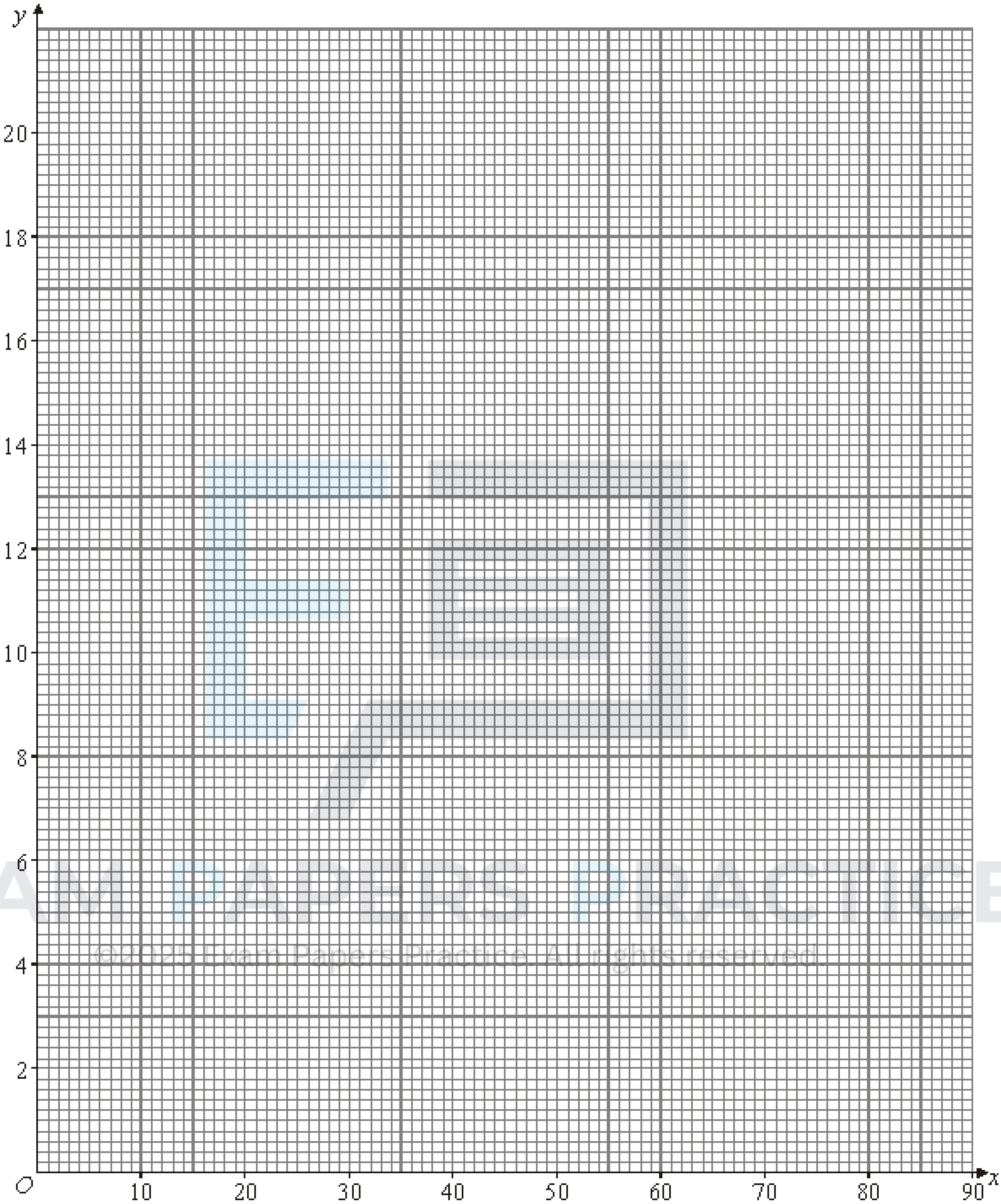
The strawberry grower has 45 kg of strawberries, 60 kg of sugar and 40 lemons.

Assume that he makes x quantities of the jam recipe and y quantities of the jelly recipe.

(a) Explain why $\frac{1}{2}x + 6y \le 45$ and write down two other similar inequalities which x and y must satisfy.

(3)

(b) Illustrate on the graph, the region representing those (x, y) satisfying $x \geq 0$, $y \geq 0$ and the three inequalities from part (a).



(4)

(c) Identical jars are used for both the jam and the jelly. Find the maximum number of jars which he might need.

(4)

(d) The grower intends to make an equal number of jars of jam and jelly. Find the maximum number of jars of each which he can make.

(4)

(Total 15 marks)

Q52.

A market trader sells sets of pans and sets of crockery. He wants the number of sets of crockery stocked to be at least twice the number of sets of pans stocked. He can fit at most 28 sets in total on his stall. Each set of pans weighs 1 kilogram, each set of crockery weighs 2 kilograms, and his stall can hold a maximum of 50 kilograms.

(a) Introduce two variables and write each of the above conditions as an inequality in those variables.

(4)

(b) Illustrate a feasible region of those points which are non-negative and satisfy the inequalities in part (a).

(5)

(c) Each set of pans makes him £2 profit and each set of crockery makes him £3 profit. How many sets of each should he stock in order to maximise his profit? Calculate this maximum profit.

(5)

(d) The trader manages to buy his pans from a different wholesaler so that his profit on a set of pans goes up to £4. How does that affect the number of sets of pans and crockery which he should stock?

(5)

(Total 19 marks)

Q53.

A chef has 1 kg of flour, 2.4 litres of milk and 15 eggs. He wishes to make pancakes and small Yorkshire puddings, and has the following ingredients lists.

<u>8 pancakes</u>	<u>6 small Yorkshire puddings</u>
100 g flour	100 g flour
300 ml milk	200 ml milk
2 eggs	1 egg

He wishes to make as many items as possible.

(a) Formulate the chef's problem as a linear programming problem.

(3)

(b) Solve the problem graphically.

(6)

(c) How will your solution be affected if an egg is dropped and broken, so that only 14 are available?

(1)

(Total 10 marks)

Q54.

[A sheet of graph paper should be provided for use in this question].

A new language school is recruiting students for its French course and other students for its German course. All the staff are able to teach both French and German. There are enough staff to provide up to 250 hours of tutorials a week. The college requires that:

at least 50% of the students study German;
there can be at most 150 students altogether;
each student of French has an individual tutorial of 1 hour a week;
and each student of German has an individual tutorial of 2 hours a week.

Let x be the number of students on the French course and let y be the number of students on the German course.

(a) Explain why the first condition is equivalent to x, < y, and express the other two conditions as linear inequalities in x and y.

(3)

(b) Illustrate on graph paper those points satisfying the three inequalities in part (a) together with $x \geq 0$ and $y \geq 0$.

(5)

(c) The college charges £1000 for each student of French and £1500 for each student of German. Find how many students it should recruit for each course in order to maximise its income. Calculate the maximum possible income.

(5)

(d) The college proposes to increase from 50% the proportion of students who study German. Find the proportion it can be increased to without reducing the college's maximum possible income as calculated in part (c).

(3)

(Total 16 marks)

Q55.

(a) Solve the following linear programming problem graphically:

Maximise $P = x + 2y$

Subject to $3x + 17y \leq 170$

$7x + 8y \leq 175.$

(5)

(b) Starting from the initial feasible solution $x = y = 0$, apply one iteration of the simplex method to the linear programming problem given in part (a). Identify the point on your graph in part (a) corresponding to the result of your iteration of the simplex method.

(5)

(c) Give the values of the slack variables in the **final** simplex solution to the problem.

(1)

(Total 11 marks)

Q56.

A company producing dining chairs and tables requires a production plan for the next month. The company has £10 000 budgeted to buy materials. Chairs each require 20 of materials and tables £100. Tables each need 15 hours of work from craftsmen and chairs each need 4 hours. There are 1950 hours of craftsman time available per month. The company sells chairs at £80 each and tables at £350. A production plan is required which maximises potential income.

(a) Formulate the problem as a linear programming problem.

(3)

(b) Use a graphical method to solve the problem.

(6)

(c) Had the simplex method been used, slack variables would have been needed.

State what those slack variables would have represented and give their values at the solution.

(2)

(d) Comment on any practical difficulty there may be with your solution to this problem.

(1)

(Total 12 marks)